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Ends of maps, I

By FRANK QUINN*

Abstract

Suppose $e: M \to X$ is a map from a (finite dimensional) manifold to a locally compact space. Conditions on e are given under which it can be extended to a proper map $e': M' \to X$ by adding boundary to M. This theorem is applied to construct a mapping cylinder neighborhood for a 1-LC ANR in a manifold, resolutions of manifold factors, block bundle approximations to approximate fibrations, local flatness of embeddings, and locally flat approximations of wild embeddings.

Introductions

Suppose M is a noncompact manifold. A *completion* of M is an embedding $M \subset M'$ such that $M' - M \subseteq \partial M'$, and M' is a compact manifold. Homotopy theoretic criteria for M to have a completion were developed by Browder, Livesay, and Levine [1], and by Siebenmann [1].

The question addressed here is the analogous one for functions. If M is a manifold and $e: M \to X$ is a continuous function, a *completion* of e is an embedding $M \subset M'$ with $M' - M \subseteq \partial M$, and an extension to $e': M' \to X$ which is proper. (A map is proper if preimages of compact sets are compact.)

The main theorem is that if X is a locally compact locally 1-connected metric space, and $e: M \to X$ is a map whose end is tame, onto, has locally constant fundamental group π satisfying $\mathrm{Wh}(\pi \times \mathbf{Z}^k) = 0$ for every k, and $\dim M \geq 6$, then e has a completion.

Precise definitions will be given in Section 1. However, here is the general idea. A *neighborhood* of the end of e is an open $U \subset M$ such that $e \mid M - U$ is proper. The end is *onto* X if each neighborhood of the end maps onto X. The end is *tame* if there are homotopies pulling M outside neighborhoods of the end, which are small when projected into X. If e has a completion then such homotopies can be obtained by pushing in a collar

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neighborhood of the boundary, so this is also a necessary condition. "Locally constant fundamental group at the end" means pretty much what it sounds like.

A good example to have in mind comes from embeddings. If $X \subset N$ is a closed embedding of an absolute neighborhood retract (ANR) in a manifold, there is a closed neighborhood $X \subset M \subset N$ with a retraction $r: M \to X$. Consider $r: M - X \to X$. This map has an end where X used to be. Neighborhoods of the end are exactly deleted neighborhoods of X in M. The end is onto if X has no interior in M. The end has trivial fundamental group exactly if the embedding is 1-LC (a commonly used condition). Finally tameness is a consequence of local Poincaré duality in M and some homotopy theory.

If $r: M - X \to X$ has a completion, r': $W \to X$, then define $\partial_1 W = W - (M - X)$. A collar of $\partial_1 W \subset W$ gives a neighborhood of $X \subset M$ homeomorphic to the mapping cylinder of $r': \partial_1 W \to X$. Conversely a mapping cylinder structure on a neighborhood of X gives a completion of the end.

The main theorem applied to this example gives the following theorem: If $X \subset M$ is a 1-LC embedding without interior of an ANR in a closed manifold of dimension $\geqq 6$, then it has a mapping cylinder neighborhood. In particular this gives a new proof that a compact finite dimensional ANR has the homotopy type of a finite complex. More general statements and previous results are given in Section 3.1.

The results of this paper were announced in Quinn [4].

Organization of the paper

We have tried to arrange the paper for the convenience of those more interested in the applications of the theorem than savoring every intricacy of the proof.

Section 1 contains definitions, and prototypical theorems: existence of completions (1.4), uniqueness up to isotopy (1.5), and a homological characterization of the tameness condition (1.7).

In Section 2, a relative version of the end theorem is given. After that is an approximate version of the end theorem, and the thin h-cobordism theorem. The thin h-cobordism theorem is the source of the uniqueness statements, and it together with the approximate end theorem and some homotopy theory from Section 5 are shown to imply the end theorem.

Section 3 is devoted to applications. Strictly speaking we use the theorems stated in Sections 1 and 2, and the homological characterizations of the latter part of Section 5. Actually the proofs are intelligible after reading Section 1 and the thin h-cobordism theorem (2.6, 2.7).

The area of the applications has been dominated until now by engulfing techniques. Except for a few 5-dimensional cases, these results can now be obtained in stronger forms by straightforward, almost routine, applications of the end theorem and related techniques. Roughly speaking, the end theorem replaces most of the engulfing. The results used in the proofs of the applications are those of this paper, some surgery theory, the theorem of Edwards [1] that cell-like manifold images with the 2-disc property are manifolds, and an "uncrumpling" theorem of Cannon, Bryant, and Lacher [1].

In brief the results are:

3.1.1 A 1-LC embedding of an ANR in a manifold of dimension $\geqq 6$ has a mapping cylinder neighborhood.

3.2.2 If $\dim X \geqq 5$, the following are equivalent: (a) X is a cell-like image of a manifold. (b) X is a manifold factor. (c) $X \times \mathbf{R}^2$ is a manifold.

3.3.2 An approximate fibration with total space a manifold of dimension $\geqq n + 5$, and base a polyhedron of dimension n, is concordant to a projection of a block bundle with topological manifold fibers (there is also a π_1 condition on the fiber).

3.4.1 If M is a manifold of dimension $\geqq 6$, and the codimension of $X \subset M$ is $\geqq 3$, X is locally flat if and only if X is a manifold factor and the embedding is 1-LC.

3.4.2 If M is a homology manifold of dimension $\geqq 6$, $M - \partial M$ is a manifold, and $\partial M \subset M$ is 1-LC and an ANR, then $\partial M \times \mathbf{R}^2$ is a manifold. Further ∂M is collared in M if and only if $\partial M \times \mathbf{R}$ is a manifold.

3.5.2 If M is a closed homology manifold, $X \subset M$ has codimension $\leqq 3$, $M - X$ is a manifold, and $X \times \mathbf{R}^k$ is locally polyhedral, for some k, then M is a manifold factor.

3.5.3 If $h: X \to M$ is a topological embedding of codimension $\geqq 3$ for a manifold factor in a manifold, and $\varepsilon > 0$, then h can be ε approximated by a locally flat embedding.

More detailed statements, and references to previous work, are given in Section 3.

Section 4 contains proofs of two of the results of Section 3 which were too lengthy to incorporate in the text.

The remainder of the paper is devoted to the proof of the end theorem. In outline it is very similar to the classical case. We begin with nested sequences of manifold neighborhoods of the end. Handles are deleted from these neighborhoods to make them highly connected. Finally when the neighborhoods are $n-3$ connected a "stable algebra" difficulty is encountered.

If this is overcome, the neighborhoods divide the end into a sequence of h-cobordisms. Application of the h-cobordism theorem then gives a collar of the end.

The major difference here is that we must keep everything small when projected to X, so that the collar finally constructed will converge in X.

Section 5 contains some ε-homotopy theory over a space X, in particular a version of the Hurewicz theorem. This is used, among other places, to find homotopy classes of handles to delete in the improvement of the neighborhoods of the end.

Section 6 begins the handlebody theory. It gives an ε version of a geometrical connectivity lemma of Wall [1]. This is used to reduce the proof of the thin h-cobordism theorem to a stable algebra problem.

Section 7 continues the handlebody theory. It employs the previous sections to reduce the approximate end theorem (roughly the outline above except for the last application of the h-cobordism theorem) to a stable algebra problem.

Finally Section 8 provides solutions to the stable algebra problems encountered in Sections 6, 7. These questions are set in the geometric group context developed by Connell and Hollingsworth [1], and the theorem can be described roughly as a solution of the conjectures advanced in that paper. The central ingredient of the proof is a Kirby torus argument, as might have been expected.

Conventions

We employ the notation $\underline{X}_f$ for the mapping cylinder of $f\colon Y \to X$. Also if X is a metric space, $K \subset X$, and $\varepsilon > 0$, then $\underline{K}^\varepsilon$ is the ε-neighborhood of K, $\{x \mid d(x, K) < \varepsilon\}$. $\underline{K}^{-\varepsilon}$ shrinks K a little; it is the closed set $X - (X - K)^\varepsilon$. $\underline{K}^0$ is the interior of K.

Historical note

The end theorem was first formulated as a conjecture in December 1969, in the course of an investigation of cell-like maps of manifolds (see Siebenmann [2], p. 276 line 12). Most of the proof given was worked out at that time except for the geometric group material of Section 8. The applications of Sections 3.1, 3.2, 3.3 were also observed to follow from the theorem.

With the advent of pseudoisotopy theory, it was suspected that there were actually nontrivial obstructions in the end problem, lying perhaps in some "Whitehead space". Attempts to formulate such obstructions were

influential in Hatcher's investigation [1].

In recent years so many of the consequences of the end theorem were proved independently that it became doubtful that there could be very many obstructions. The investigation was resumed in the fall of 1977. A proof of the theorem for 1-LC ends was obtained in early 1978 by "classifying" tame ends, and then comparing with the 1-sided taming theorem of Černavskii, Seebeck, and Ferry (Ferry [1]) to see that they all had completions.

Finally the arrival of T. A. Chapman's preprint [3] made it inconceivable that the geometric group conjectures were false. The treatment presented here followed shortly.

Since this paper was written there have been two further developments. In Part II the main theorem is extended to ends without the $\mathrm{Wh}(\pi \times \mathbf{Z}^n) = 0$ restriction. An obstruction is encountered (true to our suspicions) in the generalized homology group $H_0^{lf}(X; \mathbb{S}(\pi))$, where $\mathbb{S}(\pi)$ is a spectrum similar to Hatcher's space.

The second development is a surgery theory for ends. The simplest application is that every ANR homology manifold has a manifold resolution. Transversality is used to construct a "degree 1" end over the homology manifold, surgery is used to make it tame, and this paper provides a completion which is automatically a resolution. (See 3.4.2.)

This result makes 3.2.1 out of date (though it is used in the construction). 3.5.2 is now completely obsolete.

1. The end theorem

Suppose M is a manifold, X a locally compact space, and $e: M \to X$ is a map. A (manifold) *completion* of e is a manifold $M' \supset M$ with $M' - M \subseteq \partial M'$, and an extension of e to a proper map $e': M' \to X$. The end theorem gives conditions on e which insure the existence of a completion.

In the situation above, a *neighborhood of the end* (of e) is an open set $U \subset M$ such that $e \mid M - U$ is proper.

We recall that if X is a metric space and $\varepsilon: X \to (0, \infty)$ is continuous, a homotopy $Y \times I \to X$ is said to have *diameter* $< \varepsilon$ if the diameter of each path $\{y\} \times I \to X$ is less than the minimum of ε on the path.

1.1 *Definition.* Suppose $e: M \to X$, and X is a metric space. Then e is *tame* if for every neighborhood U of the end, and $\varepsilon: X \to (0, \infty)$ there are a neighborhood of the end $V \subset U$ and a homotopy $h: M \times I \to M$ such that $h_0 = I_M$, $h_1(M) \subset M - V$, h keeps $M - U$ inside $M - V$, and $eh: M \times I \to X$ has diameter less than ε.

If e has a completion $e'\colon M' \to X$, and $M' - M$ is a submanifold of $\partial M'$ (as will usually be the case) then the end is tame. The homotopies can be obtained by pushing along the collar coordinate in a collar of $M' - M$ in M'.

There is a homological characterization of tameness, harder to formulate but easier to verify, which will be given at the end of the section.

1.2 *Definition.* Suppose $e\colon M \to X$. Then the end of e is

1) 0-LC if for every $x \in X$, and neighborhoods $U \subset M$ of the end and $V \subset X$ of x, there are neighborhoods $U' \subset U$ of the end, and $V' \subset V$ of x such that points in $U' \cap e^{-1}(V')$ can be joined by arcs in $U \cap e^{-1}(V)$.

2) 1-LC if in addition U', V' can be chosen so that loops in $U' \cap e^{-1}(V')$ can be contracted in $U \cap e^{-1}(V)$.

3) *Onto* if for every neighborhood U of the end, $e(U) = X$.

None of these conditions are necessary for e to have a completion. However it is reasonable to impose them in view of counterexamples on the one hand, and the wide variety of applications in which they are satisfied on the other hand. We will relax the 1-LC condition somewhat. The next definition can be formulated using π_1 but the one given is more convenient for our purposes.

1.3 *Definition.* Suppose $e\colon M \to X$ is 0-LC. Then e has a *locally constant fundamental group at the end* if for each $x \in X$ there are neighborhoods V of x in X and U of the end such that $e^{-1}(V) \cap U$ has a covering space whose end is 1-LC over V.

The *local fundamental group* of the end is the group of covering transformations of this cover. It is well defined up to isomorphism, and defines a twisted local coefficient system on X.

1.4 THEOREM (*absolute end theorem*). *Suppose X is a locally compact and locally 1-connected metric space. Suppose $e\colon M \to X$ is proper on ∂M, and the end of e is onto,* 0-LC, *has locally constant fundamental group π such that* $\mathrm{Wh}(\pi \times \mathbf{Z}^k) = 0$ *for all k, and the dimension of M is $\geqq 6$. Then e has a completion $e'\colon M' \to X$. M' is a smooth or* PL *manifold if M is.*

More general end theorems will be given in the next section. This is the prototype however, and the others can be conveniently thought of as variations on this one. We remark that the $\mathrm{Wh}(\pi \times \mathbf{Z}^n)$ hypothesis can be weakened to $\tilde{K}_{-i}(\mathbf{Z}\pi) = 0$, $i \geq 0$.

The next result is the prototypical uniqueness theorem.

1.5 THEOREM. *Suppose $e\colon M \to X$ satisfies the conditions of* 1.4, *and $e'\colon M' \to X$, $e''\colon M'' \to X$ are two completions of e. Then for every $\varepsilon > 0$ there*

is an isotopy of M, with diameter $<\varepsilon$ when projected to X, fixed on some neighborhood of the end, which takes a collar neighborhood of $M' - M$ to a collar neighborhood of $M'' - M$.

This is a simple consequence of the thin h-cobordism Theorem 2.7.

Once the interior ends of these collars are identified, we can straighten out the collar parameters to get an isomorphism $M' \simeq M''$. However this moves M around near the end, which is usually undesirable.

Finally we state a homological criterion for tameness. For simplicity we assume the fundamental group is globally constant and equal to π. $H_n(-; \mathbf{Z}\pi)$ is $\mathbf{Z}\pi$ coefficient homology: homology of the 1-LC cover regarded as a $\mathbf{Z}\pi$ module. Recall also that if $K \subset X$ and $r: X \to (0, \infty)$, $K^r = \{x \in X \mid d(K, x) < r(x)\}$.

1.6 *Definition.* Suppose $e: M \to X$ is 0-LC and has constant fundamental group π at the end. The end of e is *homologically tame* if π is finitely presented, and for every neighborhood U of the end and $\varepsilon: X \to (0, \infty)$ there is a neighborhood V of the end such that for every $K \subset X$ the homomorphism

$$H_*(e^{-1}(K), e^{-1}(K) - U; \mathbf{Z}\pi) \longrightarrow H_*(e^{-1}(K^\varepsilon), e^{-1}(K^\varepsilon) - V; \mathbf{Z}\pi)$$

is zero.

1.7 PROPOSITION. *An onto, 0-LC end with constant fundamental group over a locally compact locally 1-connected metric space is tame if and only if it is homologically tame.*

A tame end is fairly clearly homologically tame. The proof of the converse, and related homological criteria, are given in Section 5.

2. Relative and approximate end theorems, and the thin h-cobordism theorem

This section consists mostly of definitions and the statements of the theorems of the title. After the statements we indicate how the last two theorems and some homotopy theory from Section 5 imply the end theorem.

Suppose $e: M \to X$ is given, and $C \subseteq X$ is open. A *completion of e on ∂M and over C* is an extension $e': M' \to X$ such that $e': (e')^{-1}(C) \to C$ is a completion of $e \mid e^{-1}(C)$ and e': (closure of ∂M) $\to X$ is a completion of $e \mid \partial M$.

An end is *tame* over C if the homotopies of Definition 1.1 are only required to exist over C: given $U \subset M$, $\varepsilon > 0$, then there are $V \subset U$ and a homotopy $h: e^{-1}(C) \times I \to M$ of $e^{-1}(C)$ into $M - V$ which keeps $M - U$ in $M - V$, and eh has diameter $<\varepsilon$.

Finally the onto, 1-LC, and locally constant fundamental group conditions are conditions on e near a point $x \in X$. We say the conditions hold over C if they hold for each point in C.

2.1 THEOREM (*relative end theorem*). *Suppose X is a locally compact and locally 1-connected metric space, and $X \supset C \supset D$ are closed. Suppose $e\colon M \to X$ is a map from a manifold of dimension $\geqq 6$ which has a completion on ∂M and over a neighborhood of D, e is tame onto, 0-LC and has locally constant fundamental group π over a neighborhood of C, and $\mathrm{Wh}(\pi \times \mathbf{Z}^k) = 0$ for all k. Then there is a completion of e on ∂M and over a neighborhood of C which agrees with the given one on ∂M and over D.*

For the rest of the section we restrict to compact X and ends with globally constant fundamental group. This significantly simplifies the definitions but does not essentially weaken the theorems.

A subset $N \subset M$ is a *manifold neighborhood* of the end of e if it is a closed codimension 0 submanifold and e is proper on the closure of $M - N$. Notice the mild inconsistency that since N is not open it is not a neighborhood of the end. We define $\underline{\partial_0 N}$ to be $N \cap$ (closure of $M - N$), and require also that $\partial_0 N$ be a submanifold of ∂N.

2.2 *Definition.* Suppose $e\colon M \to X$, M a manifold and X a metric space. Then an (ε, W_0, W_1) *approximate completion* of the end of e for $W_1 \subseteqq W_0 \subset M$ and $\varepsilon > 0$ is a manifold neighborhood of the end $M \supset N \supset W_0$, together with a homotopy of $N - W_0$ in $N - W_1$ rel $\partial_0 N$ into $\partial_0 N$, which has diameter $< \varepsilon$ in X.

Similarly an *approximate* (ε, W_0, W_1) *completion of ∂M and over C* is a neighborhood $N \supset W_0$ so that $N \cap \partial M$ is an $(\varepsilon, W_0 \cap \partial M, W_1 \cap \partial M)$ completion of $\partial M \to X$, and $e^{-1}(C) \cap (N - W_0)$ deformation retracts into $\partial_0 N \cap e^{-1}(C^\varepsilon)$ in $N - W_1$, by a homotopy of diameter $< \varepsilon$ in X.

An approximate completion can be thought of as a particularly nice form of tameness. Unfortunately the hypothesis of the theorem requires a rather unpleasant form of tameness. We begin with an idea which combines onto, 0-LC, and 1-LC.

2.3 *Definition.* Suppose X is compact metric,

$$\begin{array}{ccc} A_0 & \xrightarrow{\quad f \quad} & A_1 \\ & {\scriptstyle e_0}\searrow \quad \swarrow{\scriptstyle e_1} & \\ & X & \end{array}$$

is a commutative diagram $X \supset C$, and $\delta > 0$. Then f is $(\delta, 1)$ *connected over*

C if for every relative 2-complex (R, S) and diagram

$$\begin{array}{ccc} S & \xrightarrow{r} & A_0 \\ \cap & & \downarrow \\ R & \xrightarrow{s} & C \subseteq X \end{array}$$

there is a "δ lift of R into A_1": a map $g: R \to A_1$ such that $g \mid S = fr$, and $d(s, e_1 g) < \delta$.

A map $A \to X$ is said to be $(\delta, 1)$ connected over C if the identity map has this δ lifting property.

Now suppose $e: M \to X$ as above, $X \supset C$, and $M\hat{} \to M$ is a regular covering space with group π.

2.4 *Definition.* A (W, k) tame structure on e over C is a sequence $U_0 \supset U_1 \supset \cdots \supset U_k \supset W_0 \supset W_1 \supset \cdots W_k$ such that

1) Each U_i is a neighborhood of the end, and contains the closure of U_{i+1}.

2) The induced covers $U_j\hat{} - W_j\hat{} \to U_{j-1}\hat{} - W_{j+1}\hat{}$, and $U_j\hat{} - U_{j+1}\hat{} \to U_{j-1}\hat{} - U_{j+2}\hat{}$ are $(\delta, 1)$ connected over C', and

3) $e^{-1}(C) \cap (U_j - W_j)$ deformation retracts rel $U_j - U_{j+1}$ into $U_j - U_{j+2}$, in $U_j - W_{j+1}$ by a homotopy of diameter $< \delta$ in X.

If e already has an $(\varepsilon, W_0', W_1')$ completion N over $D \subseteq C$, we say the tame structure is *parallel to* N over D if there is a collar $\partial_0 N \times I$ *of* $\partial_0 N$ in N such that each $e^{-1}(D) \cap U_i$ is of the form $\left(N \cap e^{-1}(D)\right) - \partial_0 N \times [0, t]$, and $W_0 \supseteq W_0' \supseteq W_1' \supseteq W_k$.

2.5 THEOREM (*approximate end theorem*). *Suppose X is compact metric $X \supset C \supset D$, $\varepsilon > 0$, X is locally 1-connected in a neighborhood of C^ε, and $n \geq 6$ is given. Then there is a $\delta > 0$ and j such that if*

1) *$e: M^n \to X$ has a regular π cover $M\hat{} \to M$ with $\mathrm{Wh}(\pi \times \mathbf{Z}^k) = 0$ for all k;*

2) *e has an approximate (δ, W_0, W_1) completion on ∂M and over D^ε;*

3) *e has a (δ, j) tame structure over C^ε with respect to the cover $M\hat{}$ which is parallel to the approximate completion on ∂M and over D^ε,*

then there is an approximate (ε, W_0, W_j) completion on M and over C which agrees with the given one over D, and lies between U_0 and U_j.

This theorem will be proved in Section 7. We will see that a sufficient number of stages is 2^k where $k = (n + 1)(n + 2)/2$ in general, or $n + 2$ if $W = \varnothing$.

We remark that there is a "stable" 5-dimensional version: given the

hypothesis when $n = 5$, then the conclusion applies to an end obtained from e by connected sum along proper arcs $[0, 1) \subset M$ with $S^2 \times S^2 \times [0, 1)$.

The next theorem is the analogue of the classical h-cobordism theorem.

2.6 *Definition.* Suppose $e: M \to X$ is proper, $\partial_0 M \subset \partial M$ is a proper codimension 0 submanifold, and $\partial_1 M$ is the closure of $\partial M - \partial_0 M$.

1) $(M, \partial_0 M)$ is a (δ, h)-*cobordism over* $C \subseteq X$ if for $i = 0, 1$ there are homotopies of $e^{-1}(C)$ rel $e^{-1}(C) \cap \partial M$ into ∂M, of diameter $< \delta$ in X.

2) $(M, \partial_0 M)$ is a δ *product over* C if there is an embedding $(\partial_0 M \cap e^{-1}(C^\delta)) \times I \to M$ which is the identity on $(\partial_0 M \cap e^{-1}(C^\delta)) \times \{0\}$, which contains $e^{-1}(C)$, and which has diameter $< \delta$ in X.

2.7 THEOREM (*thin h-cobordism theorem*). *Suppose X is a compact metric space, $\varepsilon > 0$, $X \supset C \supset D$, X is locally 1-connected in a neighborhood of $C^{2\varepsilon}$ and $n \geqq 6$. Then there is $\delta > 0$ so that if $e: (M^n, \partial_0 M) \to X$ is proper, is a (δ, h)-cobordism over $C^{2\varepsilon}$, has a δ product structure over D^ε, and has a regular π cover $\hat{M} \to M$ which is $(\delta, 1)$ connected over C^ε and satisfies $\mathrm{Wh}(\pi \times \mathbf{Z}^k) = 0$ for all k, then there is an ε product structure over C which agrees with the given one over D.*

This theorem will be proved in Section 6.

Again there is a 5-dimensional version which works after connected sum with $S^2 \times S^2 \times I$ along arcs which run from $\partial_0 M$ to $\partial_1 M$ and have small diameters in X.

2.8 *Proof of the end theorem* (from 2.5 and 2.7). First, it is easy to use a patching argument to reduce the proof to the case X compact and the fundamental group of the end constant over a neighborhood of C.

Suppose ε is so small that $e: M \to X$ has a completion over D^ε, and is tame over $C^{2\varepsilon}$. Fix a collar $N \times (0, 1)$ of the end over D^ε, and on ∂M.

Choose $\varepsilon_i > 0$, $i \geqq 0$, so that $\sum_{i=0}^{\infty} \varepsilon_i < \varepsilon$. Now suppose we can find manifold neighborhoods $U_0 \supset U_1 \supset \cdots \supset U_i \supset \cdots$ of the end such that

1) U_i intersects the collar over D^{r_i} in a smaller collar, where $r_i = \sum_{j=i+1}^{\infty} \varepsilon_j$.

2) The product structure on $U_i - U_i^0$ over $D^{r_{i+1}}$ and on $\partial M \cap U_i - U_i^0$ extends to a product structure over $C^{r_{i+1}+\varepsilon}$ and on ∂M, of diameter $< \varepsilon_i$ as a homotopy in X.

3) $\bigcap U_i = \varnothing$.

Then we can put the product structures on the $U_i - U_{i+1}^0$ together to get a proper embedding $(\partial_0 U_0 \cap e^{-1}(C^\varepsilon)) \times [0, 1) \to M$ which extends the product structure on ∂M and over D. The paths $\{v\} \times [0, 1) \to X$ have diameter

$< \sum \varepsilon_i < \varepsilon$, and are Cauchy. The map $(\partial_0 U \cap e^{-1}(C^\varepsilon)) \times [0, 1) \to X$ therefore extends to $(\partial_0 U \cap e^{-1}(C^\varepsilon)) \times [0, 1] \to X$, and provides a completion of the end over C^ε.

We describe how to find such U_i. First, by virtue of the thin h-cobordism theorem, there are $\delta_i > 0$ so that it is sufficient that the U_i satisfy (1), (3) and for each $(U_i - U_{i+1}^0, \partial_0 U_i)$ to be a (δ_i, h) cobordism. Further, by 5.4 there are $\gamma_i > 0$ so that if $U_i - U_{i+1}^0$ γ_i retracts to $\partial_0 U_i$ over C^ε and $U_i^\wedge - U_{i+1}^{\wedge 0}$ and $\partial_0 U_{i+1}^\wedge$ are $(\gamma_i, 1)$ connected over C^ε, then they are (δ_i, h) cobordisms.

Next we observe that if the γ_i are decreasing, and U_i is a $((1/2)\gamma_i, \varnothing)$ approximate completion over C^ε, then these hypotheses are satisfied. Such a sequence of approximate completions can be constructed by induction. First choose a sequence $V_0 \supset V_1 \supset \cdots$ so that $\bigcap V_i = \varnothing$. Suppose we have $U_0 \supset U_1 \cdots \supset U_k$ such that U_i is a $(\gamma_i/2, \varnothing)$ approximate completion over C^ε which extends the given one on D^ε, and $U_i \subset V_i$.

According to the approximate completion theorem there are $\rho > 0$ and j so that if there is a (ρ, j) tame structure over C^ε, contained in $U_k \cap V_{k+1}$, then there is a $(\gamma_{k+1}/2, \varnothing)$ approximate completion $U_{k+1} \subset U_k \cap V_{k+1}$. The argument is completed by appealing to 5.5, which asserts that since the end is tame, onto, etc. over C^ε, such tame structures do exist.

3. Applications

Many of the applications considered here are to an area in the topology of manifolds described in a recent survey article by R.C. Lacher [1] (Cell-like mappings and their generalizations, BAMS 83 (1977), 495–552). Therefore much of our summary of previous progress on these problems is drawn from this paper, and we refer the interested reader to it for a fuller account. Also, a reference of the form Bryant [L3] refers to the bibliography of Lacher [1]. Another relevant and useful review article is that of Cannon [1].

3.1 *Mapping cylinder neighborhoods*. A mapping cylinder neighborhood of $X \subset M$ is a closed neighborhood U with frontier $\partial_0 U$, together with a map $f: \partial_0 U \to X$ and a homeomorphism $U \simeq X_f$ which is the identity on X and $\partial_0 U$. Here X_f denotes the mapping cylinder of f.

Plainly if M is a manifold and X has a mapping cylinder neighborhood, then X is an ANR. In the other direction a PL regular neighborhood of a subcomplex is a mapping cylinder (M. Cohen [L2]), a locally flat topological submanifold has one (R.D. Edwards [L4]) and finally a 1-LC embedding of

an ANR in the stable range 2 dim $X + 3 \leqq n$ has one (Miller-West-Edwards-Siebenmann; see [L, § 10]).

Our result is that this is true without the dimension restriction.

3.1.1 THEOREM. *Suppose $X \subset M$ is a closed* 1-LC *embedding without interior of an* ANR *into an n-manifold, $n \geqq 6$, and suppose $X \cap \partial M$ has a manifold mapping cylinder neighborhood in ∂M. Then it extends to a manifold mapping cylinder neighborhood of X in M which is unique up to ambient isotopy fixing ∂M and a neighborhood of X.*

We note that this uniqueness matches up the neighborhood, but not the mapping cylinder structure. Also, if the neighborhood is already given on a neighborhood of $C \subseteqq X$, then we can extend it to the rest of X.

Proof. Since X is an ANR there is a closed neighborhood N of X with a retraction $e: N \to X$ which is the mapping cylinder projection on $N \cap \partial M$. The restriction $e: N - X \to X$ has an end which is 1-LC by hypothesis, and is onto since $X \subset M$ has no interior. If this end has a completion, there is an open collar of the end which gives the desired mapping cylinder neighborhood in M. According to Theorem 1.4 (or the relative version 2.1) this end has a completion which is suitably unique (1.5) if it is tame. Finally to verify that it is tame it is sufficient to verify (by 1.7) that it is homologically tame.

Let V be a neighborhood of X in N, and $\varepsilon: X \to (0, \infty)$. Since X is an ANR, there is a neighborhood W of X in Y which deformation retracts to X in V by a retraction H such that eH has diameter $< \varepsilon$.

Now take $K \subset X$, and enlarge a little if necessary so that it is open. By Alexander duality

$$\begin{array}{ccc} H_*(e^{-1}(K) - K, e^{-1}(K) - V) & \longrightarrow & H_*(e^{-1}(K^\varepsilon) - K^\varepsilon, e^{-1}(K^\varepsilon) - W) \\ \| & & \| \\ H^{n-*}(V, X \cup e^{-1}(X - K)) & \longrightarrow & H^{n-*}(W, X \cup e^{-1}(X - K^\varepsilon)) \end{array}$$

commutes. But W deforms into X by an ε deformation, so $X \cup e^{-1}(X - K^\varepsilon)$ stays in $X \cup e^{-1}(X - K)$ during the deformation. Therefore the bottom homomorphism is zero. It follows that the top one is also, and the definition of homological tameness is verified.

3.2 *Resolutions of homology manifolds.* A *resolution* of (X, Y) is a manifold M and a cell-like map $f: (M, \partial M) \to (X, Y)$ such that $f^{-1}(Y) = \partial M$. If X is finite dimensional it follows that (X, Y) is a ANR homology manifold. The "main conjecture" of topological desingularization theory as posed in Lacher [1] is: does X have a resolution if and only if $X \times \mathbf{R}$ is a manifold?

A recent result of R.D. Edwards [1] almost gives one direction. He

shows that a resolvable homology manifold of dimension $\geqq 5$ with the "2-disc property" is in fact a manifold. Since $X \times \mathbf{R}^2$ has the 2-disc property (first observed by R. Daverman), X resolvable implies that $X \times \mathbf{R}^2$ is a manifold.

Our result is the converse, that a manifold factor has a resolution. The best previous result seems to be that of Cannon, Bryant, and Lacher [1], which gives a resolution if the non-manifold set has dimension $\leqq (1/2 \dim X) - 1$ without presuming it is a manifold factor. Results of this type will be considered in Section 3.5.

3.2.1 THEOREM. *If $X \times \mathbf{R}^n$ has a resolution of dimension $\geqq n + 5$, then X has a resolution.*

This remains true if $\partial X \neq \varnothing$ (in the obvious sense) provided either $\dim X \geqq 6$ or ∂X is a manifold. Combining this with the discussion above, we get

3.2.2 COROLLARY. *If X is a metric space, of finite dimension $\geqq 5$, the following are equivalent:*

a) *X is resolvable;*

b) *$X \times \mathbf{R}^n$ is a manifold, for some n;*

c) *$X \times \mathbf{R}^2$ is a manifold.*

Proof of 3.2.1. Suppose that $f: M \to X \times \mathbf{R}^n$ is a cell-like map. Let $e: M \to X \times \mathbf{R}^{n-1}$ be the composition with the projection. e has two ends each of which is 1-LC and tame (since f is cell-like). By the end theorem there is a completion $e': M' \to X \times \mathbf{R}^{n-1}$. It is easily seen that e' restricted to a component of $M' - M$ is cell-like, and therefore a resolution. Therefore $X \times \mathbf{R}^{n-1}$ has a resolution, and by induction X does also.

We get an interesting variation by substituting the approximate end theorem in the above proof. Define an *ε-resolution* to be a map $f: M \to X$ which has $g: X \to M$ so that fg is ε homotopic to 1_X, and gf is homotopic to 1_M by a homotopy which has diameter $< \varepsilon$ when composed with f.

The proof then yields that given X, n, and $\varepsilon > 0$, there is $\delta > 0$ so that if $X \times T^n$ has a δ resolution of dimension $\geqq n + 5$, then X has an ε-resolution. In fact there is even a (worthless) 4-dimensional result from the stable approximate 4-dimensional end theorem: if X^4 has a dense set of manifold points, $\varepsilon > 0$, n given, then there is $\delta > 0$ so that if $X \times T^n$ is resolvable, then the object obtained from X by many connected sums with $S^2 \times S^2$ on discs of diameter $< \delta$ in the manifold set is homotopy equivalent to a manifold, by a homotopy equivalence whose homotopies have diameter $< \varepsilon$ when projected into X.

Here we have used the torus T^n instead of $\mathbf{R}^n$ because we have not

formulated a noncompact version of the approximate end theorem.

A more interesting variation yields the uniqueness of resolutions. If $f: M \to X$ is a resolution and X is a manifold, then Siebenmann's theorem [L8] states that f can be ε approximated by a homeomorphism. Therefore two resolutions are (ε) homeomorphic. We extend this to the nonmanifold case, answering question 13 of Lacher [1].

3.2.3 PROPOSITION. *Suppose X is compact metric, $f_0: M_0 \to X$, $f_1: M_1 \to X$ are resolutions and* $\dim X \geqq 5$. *Then for any $\varepsilon > 0$ there is a homeomorphism $h: M_0 \to M_1$ so that $d(f_1 h, f_0) < \varepsilon$.*

Proof. According to 3.2.1, $X \times T^2$ is a manifold. If $\delta > 0$, the theorem of Siebenmann referred to above implies there is a δ-commutative diagram

$$\begin{array}{ccc} M_0 \times T^2 & \xrightarrow{h_0} & M_1 \times T^2 \\ & \searrow \quad \swarrow & \\ & X \times T^2 . & \end{array}$$

Let $\hat{h_0}$ be the map of covers induced from the second factor of T^2, e_0, e_1 the projections

$$\begin{array}{ccc} M_0 \times S^1 \times \mathbf{R} & \xrightarrow{\hat{h_0}} & M_1 \times S^1 \times \mathbf{R} \\ & \searrow^{e_0} \quad \swarrow_{e_1} & \\ & X \times S^1 . & \end{array}$$

Then $\hat{h_0}\,(M_0 \times S^1 \times \{0\})$ is a δ approximate completion of either end of e_1. Choose t so that $M_1 \times S^1 \times \{t\}$ is disjoint from this image; then the region between them is a thin h-cobordism (see 2.6, 5.4). It is $(\delta, 1)$ connected since the maps are obtained from cell-like maps. Therefore according to the thin h-cobordism Theorem (2.7) if $\gamma > 0$ there is $\delta > 0$ so that it is a γ product. In particular there are a homeomorphism h_1 and a γ commutative diagram

$$\begin{array}{ccc} M_0 \times S^1 & \xrightarrow{h_1} & M_1 \times S^1 \\ & \searrow \quad \swarrow & \\ & X \times S^1 . & \end{array}$$

Repeat this argument to obtain the homeomorphism of the proposition.

This proposition extends to the bounded case if either $f_i: \partial M_i \to \partial X$ is a homeomorphism (and then $h \mid \partial M_0 = f_1^{-1} f_0$), or if $\dim X \geqq 6$.

3.3 *Approximate fibrations.* A *fibration* is a map with homotopy lifting property: $f: E \to B$ so that given a commutative diagram

$$\begin{array}{ccc} K \times \{0\} & \xrightarrow{H_0} & E \\ \cap & & \downarrow f \\ K \times I & \xrightarrow{h} & B \end{array}$$

there is $H\colon K \times I \to E$ with $fH = h$ and $H \mid K \times \{0\} = H_0$.

If B is a metric space, we can weaken this to (ε, n) *fibration* by requiring that if K is a polyhedron of dimension $\leqq n$, there is an extension H of H_0 such that $d(fH, h) < \varepsilon$. Finally f is an *approximate fibration* if it is an (ε, n) fibration for each $\varepsilon > 0$, and each n.

This concept has been developed by Coram and Duvall [L1, 2] and Goad [L1] among others. A principal question in the area has been: can an approximate fibration between manifolds be approximated by a fibration? Some slight progress has been made in this direction by the above and Chapman [1]. This turns out to be slightly the wrong question, as we shall see. The first lemma provides a supply of examples.

3.3.1 LEMMA. *Suppose K is a polyhedron, and $f\colon E \to K$ is a topological block bundle.*

1) *f is homotopic through block bundle maps to $f'\colon E \to K$ which is also an approximate fibration.*

2) *If the fiber F is a topological manifold of dimension $\geqq 1$ ($\geqq 2$ if $\partial F \neq \varnothing$) and K is a homology manifold, then E is a topological manifold.*

Proof. A topological block bundle with fiber F may be conveniently thought of as a map $f\colon E \to K$ and a triangulation of K such that for every simplex σ in K there is a homeomorphism $f^{-1}(\sigma) \simeq F \times \sigma$, which carries $f^{-1}(\partial_i \sigma)$ to $F \times \partial_i \sigma$.

We prove (1) by induction: Suppose f is an approximate fibration over the k skeleton of K (obviously true for $k = 0$). Let σ be a $k+1$ simplex, and choose a homeomorphism $f^{-1}(\sigma) \simeq F \times \sigma$. Consider σ as the cone on its boundary $c(\partial\sigma)$, and define a new map $F \times \sigma \to c(\partial\sigma)$ by: on a collar of $\partial\sigma$ define $F \times \partial\sigma \times I \to c(\partial\sigma)$ by $f \mid f^{-1}(\partial\sigma)$ on the $F \times \partial\sigma$ coordinate, and take the collar coordinate to the cone coordinate. Map the complement of the collar to the cone point. This map agrees with f on $F \times \partial\sigma$, so is homotopic to f rel $F \times \partial\sigma$ since σ is contractible.

It is easy to use the homotopy to construct a concordance which is constant on the k skeleton to a map which has the new map over each $k+1$ simplex. This is an approximate fibration over the $k+1$ skeleton.

In assertion (2) suppose $x \in E$, and $f(x)$ lies in the interior of a simplex σ. Then x has a neighborhood of the form $c(\text{link}(\sigma, K)) \times \sigma \times \mathbf{R}^j$, $j = \dim F$.

Therefore it is sufficient to show that if Σ is a PL homology manifold which is a homology sphere, then the interior of $(c\Sigma) \times \mathbf{R}$ is a manifold. Since this has the disjoint 2-disc property, and any such Σ is PL resolvable (Cohen-Sullivan [L1], [L2]) this fact follows from Edwards' theorem [1].

This completes the proof of the lemma. The main theorem of the section is essentially a converse to the lemma.

3.3.2 THEOREM. *Suppose M is a topological manifold without boundary, K a polyhedron, and $f: M \to K$ a proper approximate fibration. If the "fiber" is connected, has fundamental group π satisfying* $\mathrm{Wh}(\pi \times \mathbf{Z}^k) = 0$ *for all k, and has dimension $\geqq 5$, then for any $\varepsilon > 0$ there is an approximate fibration $H: M \times I \to K \times I$ such that $H: M \times \{0\}$ is f, $H: M \times \{1\} \to K \times \{1\}$ is a topological block bundle, and H has diameter $< \varepsilon$ as a homotopy when projected into K.*

We remark that K is automatically a homology manifold if there is such an approximate fibration; let U be the open star of a point in K. The duality $\cap [M]: H^*(f^{-1}(U)) \xrightarrow{\simeq} H^{lf}_{n-*}(f^{-1}(U))$ gives a morphism of the Leray spectral sequences for cohomology and homology of $f^{-1}(U) \to U$. The comparision theorem shows that it is an isomorphism on E_2, and in particular $H^*(U) \simeq H^{lf}_{j-*}(U)$ for some j. Since $H^{lf}_{*}(U)$ is H_{*-1} of the link of the point, the link must be a homology sphere. This sort of argument is used in the author's thesis.

Variations

1) (with boundary) If we allow boundary, $\partial M = \partial_0 M \cup \partial_1 M$ with $f^{-1}(\partial K) = \partial_0 M$, and assume f is already a block bundle on $\partial_0 M \to \partial K$ and $\partial_1 M \to K$, then we get H as above which is also a block bundle concordance on $\partial_0 M$ and $\partial_1 M$.

2) (limit of block bundle) We can in fact find H so that $H | M \times (0, 1] \to K \times (0, 1]$ is a topological block bundle.

3) ((ε, n) fibrations) If K is compact and m and ε are given, there is $\delta > 0$ so that if $f: M^m \to K$ is a (δ, m) fibration with the above restrictions on the fiber, then there is an H which is an $(\varepsilon, m + 1)$ fibration with $H | M \times \{1\}$ a block bundle.

4) (sphere bundles) If $f: M \to K$ has $f^{-1}(\partial K) = \partial M$, $\partial M \to \partial K$ is an S^j block bundle, and f is an approximate fibration with homotopy fiber S^j, then we can relax the dimension restriction to $j \geqq 1$, $m \geqq 6$.

3.3.3 *Remarks.* The proof is a bit long, so it is deferred to the next section. It is a straightforward handlebody and induction on dim K argument.

The longest part of the argument is some surgery theory needed to obtain the result for arbitrary PL homology manifolds, rather than just PL manifolds.

The proof we give actually proves variation (1). Variation (2) follows from (1): approximate f by a sequence of block bundle maps, and use the relative version to fill in concordances between them. Variation (3) follows by replacing the end theorem by the approximate end theorem in the proof.

Variation (4) requires a different proof since some of the dimensions are too low to begin the induction. We sketch a proof using a torus argument and the stability theorem for topological block bundles.

The proof can easily be modified to yield a proof of Siebenmann's theorem [L8] that a cell-like map of manifolds can be approximated by homeomorphisms. Variation (3) corresponds to the Chapman-Ferry result [1] on ε homotopy equivalences.

Finally we return to the original question: when is an approximate fibration near a fibration? According to the theorem, it is the same problem as approximating a block bundle by a fibration. This is a well known question which only rarely has a positive answer. It is possible essentially only if the fiber is S^1, or is simply connected and K has dimension 1 (this is the pseudoisotopy theorem of Cerf).

3.4 *Local flatness.* A subset $X \subset Y$ is *locally flat* if each $x \in X$ has a neighborhood U in Y such that $(U, U \cap X) \simeq ((U \cap X) \times \mathbf{R}^k, (U \cap X) \times \{0\})$. We divide the discussion into codimension $\geqq 2$, and codimension 1.

Codimension $\geqq 2$. Suppose $X \subset M$ is locally flat, and $M - X$ is a manifold. Then we conclude

a) M is a manifold;

b) X is a manifold factor;

c) $X \subset M$ is 1-LC if the codimension is $\geqq 3$, and is locally homotopically unknotted (Rushing [L1]) if the codimension is 2.

The theorem is essentially a converse. (A homology manifold is presumed to be a finite dimensional ANR.)

3.4.1 THEOREM. *Suppose $M \supset X$, M is a homology manifold of dimension $\geqq 6$, $M - X$ is a manifold, X is a manifold factor, $X \cap \partial M = \partial X$ and ∂X is locally flat in ∂M, and $X \subset M$ is* 1-LC *if the codimension is > 2, locally homotopically unknotted if the codimension $= 2$. Then X is locally flat in M.*

In codimension 2 this theorem has been proved by Chapman [2] and Seebeck, assuming that X is a manifold. Their proof includes dim $M = 5$,

however. Chapman's proof seems to apply if $X \times \mathbf{R}$ is a manifold.

In codimension $\geqq 3$ this is due essentially to Bryant and Seebeck [L5] provided both X and M are manifolds. Cannon, Bryant, and Lacher [1] are able to conclude that M is a manifold provided $2(\dim X) + 2 \leqq \dim M$, without even requiring that X be a homology manifold. Finally we mention the work of Bryant and Lacher [L2] in which it is shown that the 1-LC condition can sometimes be deduced from other data.

The proof is deferred to the next section. It involves three steps: a mapping cylinder characterization of local flatness, a destabilization result ($X \times \mathbf{R} \subset M \times \mathbf{R}$ locally flat $\Rightarrow X \subset M$ locally flat), and then application of the block bundle results of 3.3 to prove the stable result.

Codimension 1.

3.4.2 THEOREM. *Suppose M is a homology manifold of dimension $\geqq 6$, $M - \partial M$ is a manifold, ∂M is an* ANR *and $\partial M \subset M$ is* 1-LC. *Then ∂M is a resolvable homology manifold, and is collared in M if and only if $\partial M \times \mathbf{R}$ is a manifold.*

We recall that Edwards' result implies that since ∂M is resolvable, $\partial M \times \mathbf{R}^2$ is a manifold. As yet it is unknown if this implies that $\partial M \times \mathbf{R}$ is a manifold.

If we assume ∂M is a manifold, the conclusion is equivalent to: M is a manifold. This result was announced by A. V. Černavskii, and proved by C. L. Seebeck. A proof appears in Ferry [1].

A somewhat weaker 2-sided version is: suppose $X \subset M$ is a codimension 1 embedding in the interior of a manifold. If X is a manifold and 1-LC on each side, the local flatness is due to Černavskii, Daverman, Price and Seebeck (see Daverman's survey article [1]). When X is a homology manifold Cannon [L3] has shown that $X \times \mathbf{R}$ is resolvable.

Proof of 3.4.2. Since ∂M is an ANR, a neighborhood U in M retracts to ∂M by $V\colon U \to \partial M$. The map $V\colon U - \partial M \to \partial M$ has an end which is 1-LC by hypothesis, onto by homological arguments, and homologically tame by duality (see the proof of 3.1.1). Therefore ∂M has a mapping cylinder neighborhood in M; let $f\colon N \to \partial M$ be the map.

It follows from a simple homology argument that f is a cell-like map; so ∂M is resolvable. Therefore $f \times 1\colon N \times (0, 1) \to \partial M \times (0, 1)$ is also cell-like. If $\partial M \times (0, 1)$ is a manifold, we can apply Siebenmann's theorem [L8] (or Edwards' [1]) to approximate $f \times 1$ by a homeomorphism $h\colon N \times (0, 1) \to \partial M \times (0, 1)$, such that on $N \times (0, 1/i)$, the distance from h and $f \times 1$ is $\leqq 1/i$. This homeomorphism extends to a homeomorphism of the open mapping

cylinder $(\partial M)_f^0 = N \times (0, 1) \cup \partial M \to \partial M \times [0, 1)$ by the identity on ∂M. Since the mapping cylinder is a neighborhood of ∂M in M, it follows that ∂M is collared in M.

The converse, that if ∂M is collared then $\partial M \times \mathbf{R}$ is a manifold, is clear since $M - \partial M$ is a manifold.

3.4.3 *Block neighborhoods of polyhedra.* A related question is: if $K \subset M$ is a 1-LC embedding of a PL homology manifold, does it have a normal topological block bundle? The answer, using techniques similar to 3.4.1, can be seen to be yes if $\dim M \geqq 6$. However there is a more general result: if $K \subset M$ is a 1-LC embedding of a polyhedron in a manifold of codimension $\geqq 5$, then it has a mapping cylinder neighborhood whose map is a projection of manifold "blocks" to the simplices of K. This can be seen by using the argument in the proof of 3.3.2 and the dual cone complex structure of K (see Akin [1]).

3.5 *Taming.* Our object here is to start with a "wild" object (a homology manifold, a wild embedding) and approximate it by a "tame" one (a manifold, or a locally flat embedding). The results follow from a combination of a theorem of Cannon, Bryant, and Lacher which makes things 1-LC, and results earlier in this section that 1-LC things are "tame".

The following is implicit in Cannon, Bryant and Lacher [1], Sections 1-4:

3.5.1 THEOREM (*Cannon, Bryant, and Lacher*). *Suppose M is a homology manifold of dimension $\geqq 5$, $X \subset M$, and $M - X$ is a manifold. Suppose also either*

(a) *$X \subset \partial M$, or*

(b) *$X \subset M - \partial M$, and $\dim X \leqq \dim M - 3$.*

Then there are a homology manifold N and a cell-like map $f: N \to M$ which is a homeomorphism on $f^{-1}(X) \cup \partial N$, such that $N - f^{-1}(X)$ is a manifold, and $f^{-1}(X) \subset N$ is 1-LC.

3.5.2 THEOREM. *Suppose M is a homology manifold, $X \subset M$ and $M - X$ is a manifold. Suppose either*

(a) *$X \subset \partial M$, or*

(b) *$X \subset M - \partial M$ and* (1) *$\dim X \leqq \dim M - 3$,* (2) *$X \times \mathbf{R}^k$ is locally homeomorphic to polyhedra, for some k.*

Then M is a manifold factor (*hence resolvable if $\dim M \geqq 6$ (case* a) *or $\dim M \geqq 5$ (case* b)).

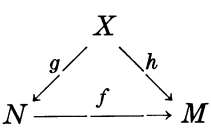
Proof. Apply the Cannon-Bryant-Lacher theorem to $X \times R^k \subset M \times \mathbf{R}^k$ to get a cell-like map $N \to M \times \mathbf{R}^k$. In the first case restrict to a neighborhood of a point in $X \times \mathbf{R}^k \subset N$ to get $W \supset K$, K a polyhedron properly 1-LC embedded, W a homology manifold, $W - K$ a manifold, and $\dim K < \dim W - 3$. W is in fact a manifold: suppose the complement of the k skeleton $W - K_{(k)}$ is a manifold. Then $K_{(k)} - K_{(k-1)} \subset W - K_{(k-1)}$ is a collection of discs which satisfy the hypothesis of 3.4.1, so in particular $W - K_{(k-1)}$ is a manifold. By induction W is a manifold. Therefore $M \times \mathbf{R}^k$ is resolvable, and by 3.2.2, M is a manifold factor.

In the second case $\partial N \subset N$ is 1-LC and $N - \partial N$ is a manifold. We conclude by 3.4.2 that ∂N is resolvable, so $\partial(N \times \mathbf{R}^2)$ is a manifold. Therefore 3.4.2 implies that $N \times \mathbf{R}^2$ is a manifold. Again by 3.2.2 this implies that M is a manifold factor.

Finally we apply this to obtain an ε-approximation theorem for embeddings of homology manifolds. The reader should consult Rushing's book [L1] and Daverman's survey article [1] for the literature on embeddings.

3.5.3 THEOREM. *Suppose $h: X \to M$ is a proper embedding of a manifold factor in a manifold M, $h(X) \cap \partial M = \partial X$, and ∂X is locally flat in ∂M. If $\dim M \geqq 6$, $\dim X \leqq \dim M - 3$, and $\varepsilon > 0$, then h can be ε approximated* rel ∂X *by a locally flat embedding.*

Proof. Apply the Cannon-Bryant-Lacher theorem to $h(X) \subset M$ to factor h:



satisfying the conclusions of 3.5.1. By 3.4.1, N is a manifold and $g(X)$ is locally flat in N. Finally apply Siebenmann's theorem to ε-approximate f rel ∂N by a homeomorphism. The composition of g and this approximation is the desired approximation of h.

3.5.4 THEOREM. *Suppose $h: X \to M$ is a codimension 1 embedding of a homology manifold in a manifold M, such that $h(X) \cap \partial M = \partial X$, and ∂X is locally flat in ∂M. Suppose also that X is 2-sided in a neighborhood in M, and that $\dim M \geqq 6$. Then h can be ε approximated* rel ∂X *by a locally flat embedding on one side of $h(X)$ if and only if $X \times \mathbf{R}$ is a manifold.*

We note that it follows from the other hypotheses that $X \times \mathbf{R}^2$ is a manifold; so the $X \times \mathbf{R}^1$ hypothesis may be redundant.

Proof. Consider one side of $h(X)$. This is a homology manifold which is a manifold in the complement of $h(X)$. Apply the Cannon-Bryant-Lacher theorem to factor X through a resolution in which it is 1-LC. Apply 3.4.2 to conclude that X is resolvable, and is collared in the resolution if and only if $X \times \mathbf{R}$ is a manifold. Apply Siebenmann's theorem to ε-approximate the resolution in the complement of X by a homeomorphism. The image of the open collar gives the desired approximation.

4. Proofs of applications

Proof of 3.3.2. If K is a PL homology manifold, a "handle decomposition" on K can be constructed as follows: triangulate, take the second barycentric subdivision, and consider relative regular neighborhoods of the "central" part of each simplex. The "central part" is the closure of the complement of the star of the boundary. This displays K as built up of handles of the form $(c\Sigma \times D^k, \Sigma \times D^k)$ where Σ is a homology sphere.

Now suppose $f\colon (M, \partial_0 M) \to (K, \partial K)$ is an approximate fibration as in 3.3.2, variation 1, and suppose $(c\Sigma \times D^k, \Sigma \times D^k) \subset (K, \partial K)$ is one of these "handles". The first step is to improve $f^{-1}(c\Sigma \times D^k)$.

Consider $M \times I - f^{-1}(c\Sigma \times D^k) \times \{1\} \to K$. The boundary $M \times \{1\} - f^{-1}(c\Sigma \times D^k) \times \{1\}$ has an end over $c\Sigma \times S^{k-1}$. More precisely take a collar of $c\Sigma \times S^{k-1}$ in K outside the handle, take the part of the manifold lying over the collar, and project to $c\Sigma \times S^{k-1}$. Since the map is a proper approximate fibration this has a tame end. Further since f is over ∂K and $f|\partial_1 M$ is a block bundle, this end has a completion on the boundary. The other hypotheses required for application of the relative end Theorem 2.1 are assumed in the hypotheses of 3.3.2, so there is a manifold completion of this end.

A similar argument shows that this extends to a completion of $M \times I - f^{-1}(c\Sigma \times D^k) \times \{1\} \to K$. By the s-cobordism theorem the manifold thus produced is $M \times I$. The map is a concordance from f to an approximate fibration such that the inverse image of $c\Sigma \times S^{k-1}$ is a codimension 1 submanifold, and the restriction $f^{-1}(c\Sigma \times S^{k-1}) \to c\Sigma \times S^{k-1}$ is an approximate fibration which is a block bundle over $\Sigma \times S^{k-1}$ and on the boundary.

This restriction satisfies the hypotheses of the theorem, except that the dimension of the total space is reduced by one. If we assume as an induction hypothesis that the theorem is true in lower dimensions, then there is a concordance of this restriction to a block bundle. This concordance can be used to construct a concordance of f to a map which is a block bundle over $c\Sigma \times S^{k-1}$.

We now have an approximate fibration on $c\Sigma \times D^k$ which is a block

bundle on the boundary. If there is a concordance rel boundary to a block bundle, then the concordance required for the theorem has been constructed over a handle, and induction on handles completes the proof. The theorem is therefore reduced to the following special case.

4.1 LEMMA. *Suppose* $f\colon (M, \partial_0 M) \to (c\Sigma, \Sigma)$ *is an approximate fibration which satisfies the conditions of* 3.3.2 *variation* 1, *and* Σ *is a homology manifold which is a homology sphere. Then* f *is concordant* rel ∂M *to a block bundle.*

Proof. We remark that if K is a PL manifold, i.e., $\Sigma = S^{k-1}$, then there is a simple direct proof. After a concordance we can use the triangulation $(\Delta^k, \partial\Delta^k) \approx (D^k, S^{k-1})$. The only thing to be verified is that $M \simeq \Delta^k \times F$, but this follows from the s-cobordism theorem applied to $(M, f^{-1}(\Delta^k - \operatorname{int} \partial_i \Delta^k))$.

The general case seems to require some surgery. First note that the inclusion $\partial_0 M \subset M$ gives a homotopy trivialization for the block bundle $\partial_0 M \to \Sigma$, which is a trivialization on $\partial_{0_1} M$ since $\partial_1 M \to c\Sigma$ is a block bundle. Let the fiber be $(F, \partial F)$; then the bundle has a classifying map $\Sigma \to G(F, \partial F)/\mathrm{TOP}^{\sim}(F, \partial F)$, where G is the monoid of self-equivalences which are the identity on ∂F, and $\mathrm{TOP}^{\sim}$ is the Δ-set of block homeomorphisms which are the identity on ∂F. This map goes into the identity component.

Since $\dim F \geqq 5$, the identity component of $G(F, \partial F)/\mathrm{TOP}^{\sim}(F, \partial F)$ can be identified with the structure set $S_{\mathrm{TOP}}(F, \partial F)$ (see Quinn [1], or Wall [2, § 17A]). The structure set, however, can be given the structure of an infinite loop space, so the identity component of $G(F, \partial F)/\mathrm{TOP}^{\sim}(F, \partial F)$ has an infinite loop space structure.

Now let H denote $c\Sigma$ minus the interior of a disc in a top dimensional simplex. (H, Σ) is acyclic. As is well known, maps to infinite loop spaces extend over acyclic pairs, so the classifying map above extends to $H \to G(F, \partial F)/\mathrm{TOP}^{\sim}(F, \partial F)$. By 3.3.1 the total space of this bundle is a manifold.

Put the original f over $c\Sigma \times \{0\} \subset c\Sigma \times I$, put the new bundle over $H \times \{1\}$, and $\partial_0 f \times I$ over $\Sigma \times I$. The total space is a manifold homotopy equivalent to $(F, \partial F) \times c\Sigma \cup_{\Sigma} H$, and so determines a map $[F, \partial F] \times [c\Sigma \times \{0\} \cup \Sigma \times I \cup H \times \{I\}] \to G/\mathrm{TOP}$. Since the pair $(c\Sigma \times I, c\Sigma \times \{0\} \cup \Sigma \times I \cup H \times \{1\})$ is cyclic and G/TOP is simply connected (or an infinite loop space as above) the map extends to $(F, \partial F) \times (c\Sigma \times I) \to G/\mathrm{TOP}$. This determines a manifold and a normal map extending the homotopy equivalence defined above. Since a part of the boundary $(c\Sigma - H)$ is left free, and since the dimension is $\geqq 5$, the fundamental theorem of surgery (Wall [2 § 4]) states that there is a homotopy equivalence $(N, \partial_1 N) \to (F, \partial F) \times (c\Sigma \times I)$ extending

the one given above.

The piece of $\partial_1 N$ over $(\Sigma - H) \times \{1\}$ is, by the s-cobordism theorem, homeomorphic to $F \times$ (disc). Since $(\Sigma - H)$ is contained in the interior of a top dimensional simplex, and the map is a block bundle outside of it, after a little adjustment of triangulations the map will be a block bundle over all of $\Sigma \times I \cup c\Sigma \times \{I\}$.

Change the projection so that it is also an approximate fibration (3.3.1). The manifold N is homotopy equivalent to F, so if we extend the map $\partial_1 N \to \Sigma \times I \cup c\Sigma \times \partial I$ into the interior by product with a collar parameter going to the cone parameter (here we write $c\Sigma \times I \simeq c(\Sigma \times I \cup c\Sigma \times \partial I)$), and map the complement of the collar to the cone point, then the result is an approximate fibration. This gives the required concordance to a block bundle, and completes the proof of 4.1.

This completes the proof of 3.3.2, and variations 1 and 2. In variation 3 the same proof works with a little more care about distances, when the approximate end theorem is substituted for the end theorem. About this we only remark that if $f: M^n \to X \times (0, 1)$ is a proper (δ, n) fibration, then $f: M \to X$ has two ends which can be given δ tameness structures. For example $U_i = f^{-1}(X \times (1/2 + i\delta, 1))$, $W_i = f^{-1}(3/4 + i\delta, 1)$ gives a (δ, k) tame structure if $k\delta < 1/4$.

Finally we come to variation 4, sphere bundles. The proof given above works except for a few low dimensional cases needed to start the induction. We sketch a proof using surgery and a torus argument which does sphere bundles in all dimensions.

Suppose $f: (M, \partial_0 M) \to (c\Sigma \times D^k, \Sigma \times D^k)$ is an approximate fibration, which is a block bundle over $\Sigma \times D^k$. Since $M \sim S^j$, these bundles are homotopically trivial. The first claim is that the homotopy trivialization over $\Sigma \times D^k$ is homotopic to a block trivialization. If $j = 1$ this is easy since a homotopically trivial S^1 bundle is trivial. If $j > 1$ the bundle has a classifying map $\Sigma \times D^k \to G(S^j)/\mathrm{TOP}^{\sim}(S^j)$ which is stably trivial, since it stably agrees with part of the normal invariant of $\partial_0 M \to \Sigma \times D^k \times S^j$ which extends to $M \to c\Sigma \times D^k \times S^j$. The claim therefore follows from the stability theorem for block bundles; $G(S^j)/\mathrm{TOP}^{\sim}(S^j) \to G/\mathrm{TOP}$ is a homotopy equivalence for all $j \geqq 2$.

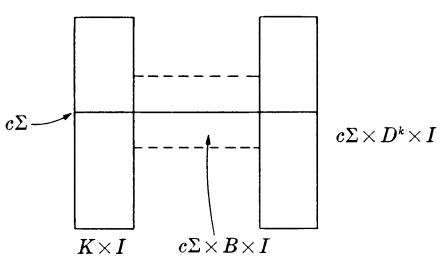
The next step is to transfer to a torus. Choose $T^{k-1} \times (0, 1) \subset D^k$. Since the total dimension is $\geqq 6$ we can use the end theorem to complete the ends of $f^{-1}(c\Sigma \times T^{k-1} \times (0, 1)) \to c\Sigma \times T^{k-1} \times (0, 1)$ and then double the result along the new boundary. This gives an approximate fibration $P \to c\Sigma \times T^k$ which is a block bundle over $\Sigma \times T^k$.

By the first step, the bundle on the boundary is trivial. Therefore there is a homotopy equivalence $P \simeq c\Sigma \times T^k \times S^j$ which is a homeomorphism on the boundary. The normal invariant of this homotopy equivalence can be shown to be trivial, partly from the construction and partly because it is detected by the $\mathbf{Z}[\pi_1 S^j \times \mathbf{Z}^k]$ surgery obstruction. Finally a homotopy equivalence with trivial normal invariant and fundamental group free abelian is homotopic to a homeomorphism (see Wall [2] §15).

Using this homeomorphism to identify P and $c\Sigma \times T^k \times S^j$, we have an approximate fibration $c\Sigma \times T^k \times S^j \to c\Sigma \times T^k$ which is the projection on the boundary $\Sigma \times T^k \times S^j \to \Sigma \times T^k$, and which is homotopic to the projection. Push in the cone parameter until it is a projection except over $c \times T^k$. Pass to the universal cover of T^k, to get $c\Sigma \times \mathbf{R}^k \times S^j \to c\Sigma \times \mathbf{R}^k$. Consider $\mathbf{R}^k$ as the interior of D^k; then this map extends to $c\Sigma \times D^k \times S^j \to c\Sigma \times D^k$ by adding the projection $c\Sigma \times S^{k-1} \times S^j \to c\Sigma \times S^{k-1}$ on the boundary.

We now have an approximate fibration $c\Sigma \times D^k \times S^j \to c\Sigma \times D^k$ which is a block bundle over the boundary. According to Lemma 4.1 it is concordant rel boundary to a block bundle.

The last step is to assemble all this to get back to the original approximate fibration. Choose an open ball $B \subset D^k$ inside the embedding of $T^{k-1} \times (0, 1)$. Then $c\Sigma \times B$ lifts to $c\Sigma \times B \subset c\Sigma \times \mathbf{R}^k \subset c\Sigma \times D^k$, and the lifted approximate fibration agrees with the original one over this set. Now take $K \times I$, attach to $K \times \{1\}$ a copy of $c\Sigma \times B \times I$, and attach $c\Sigma \times D^k \times I$ to the other end of this.



Over these pieces put $f \times I$, $(f \mid c\Sigma \times B) \times I$, and the concordance to a block bundle respectively. One last application of the end theorem fills in the map over the missing boundary ($c\Sigma \times$ (boundary of B) $\times I$). The result is a concordance to an approximate fibration which is a block bundle on $\partial K \cup \subset \Sigma \times D^k$.

As before, the proof is concluded by induction on handles.

Proof of 3.4.1. As indicated in Section 3, the proof proceeds in 3 steps, the first of which is a characterization of local flatness. The basic idea is

that if we have a mapping cylinder neighborhood $f: N \to X$, and a family of homeomorphisms $h_t: X \times S^j \to N$, $0 < t \leqq 1$, such that $d(p, fh_t) < t$ (here $p: X \times S^j \to X$ is the projection), then this extends by the identity on X to $H: X \times D^{j+1} \to X_f$. While this idea works in codimension $\geqq 3$ we are not prepared to prove it here. Instead we use a "block" version.

4.2 LEMMA. *Suppose* $f: N \to X$, N *a manifold of dimension* $\geqq 5$, X *metric, and* $x \in X$ *such that* $f^{-1}(x) \cap \partial N = \varnothing$. *Then* X *is locally flat in the mapping cylinder* X_f *at* x *if and only if there is a neighborhood* U *of* x *in* X *such that for every* i *there is an open embedding* $H_i: U \times S^j \times I \to N \times I$ *containing* $f^{-1}(U^{-1/i}) \times I$ *such that*

1) $d(p, fH_i) < 1/i$,

2) $H_i | U \times S^j \times \{0\} = H_{i+1} | U \times S^j \times \{1\}$.

The p in condition (1) is the projection $U \times S^j \times I \to U$. We actually show that given any sequence $\varepsilon_i > 0$, $\varepsilon_i \to 0$, we can find H_i satisfying the above with (1) strengthened to $d(p, fH_1) < \varepsilon_i$.

Proof. One direction is clear, since if we put H_i on $U \times S^j \times [1/i, 1/i + 1] \to N \times [1/i, 1/i + 1]$, then this extends by the identity on U to give a homeomorphism on a neighborhood of X; $U \times D^{j+1} \to N \times [0, 1]/\{(n, 0) \sim f(n)\} = X_f$.

The converse is an elaboration on the uniqueness theorem for completions. Given δ, we can find nested neighborhoods $N \times [0, r]/\sim \supset h(U \times tD^{j+1})$ so that the region between them (over $V \subset U$) δ retracts to the outer edge. It follows from 5.4 that this region is a (γ, h)-cobordism for some γ, and by the thin h-cobordism theorem if δ (hence γ) is small enough the region is an ε product. The product structure gives an ε isotopy supported near the region so that the levels $N \times \{r\}$, $U \times S^j \times \{t\}$ agree (over $V \subset U$). Choose $\varepsilon_i \to 0$, and do this on a sequence of nested neighborhoods to get ε_i isotopies with disjoint supports. These converge to an isotopy of $V \times D^{j+1} \to X_f$ which is constant on $V \times \{0\}$ to an embedding which matches up a sequence of levels in the two structures which converge to X. The embeddings on the regions between these levels provide the maps H_i.

4.3 LEMMA. *If* $X \subset M$ *satisfies the hypotheses of* 3.4.1 *and* $X \times \mathbf{R} \subset M \times \mathbf{R}$ *is locally flat, then* $X \subset M$ *is locally flat.*

Proof. Let $U \times (-t, t)$ be a neighborhood of $(x, 0) \in X \times \mathbf{R}$ over which the embedding is locally flat. Let $f: N \to X$ be the map of a mapping cylinder neighborhood of X in M. Then $f \times 1: N \times \mathbf{R} \to X \times \mathbf{R}$ is a mapping cylinder neighborhood of $X \times \mathbf{R}$ in $M \times R$.

According to the characterization of 4.2, if $\delta_i > 0$, $\delta_i \to 0$ then there is a sequence of open embeddings $H_i\colon U \times (-t, t) \times S^j \times I \to (N \times \mathbf{R}) \times I$ satisfying in particular $d(p, (f \times 1)H_i) < \delta_i$. If $2\delta_i < t$, we get inclusions

$$H_i(U^{-2\delta_i} \times \{0\} \times S^j \times I) \subset (f \times 1)^{-1}(U^{-\delta_i} \times (-\delta_i, \delta_i)) \subset H_i(U \times (-2\delta_i, 2\delta_i) \times S^j \times I) .$$

Therefore $H_i(U \times [0, t) \times S^j \times I) - (f \times 1)^{-1}(X \times (\delta_i, \infty))$, together with the map to X given by restriction of $f \times 1$ and then projection, is a (δ_i, h) cobordism over $U^{-2\delta_i}$.

First suppose $\gamma_i > 0$ and that δ_i is small enough that the boundary (δ_i, h) cobordism

$$H_i(U \times [0, t) \times S^j \times \{0\}) - (f \times 1)^{-1}(X \times (\delta_i, \infty))$$

is a γ_i product. This gives a product structure on the upper and lower boundaries of

$$H_i(U \times [0, t) \times S^j \times I) - (f \times 1)^{-1}(X \times (\delta_i, \infty)) .$$

Again by the thin h-cobordism theorem there is a γ_i small enough that these extend to an ε_i product structure over $U^{-\varepsilon_i}$.

These product structures give isomorphisms between the ends, $U \times \{0\} \times S^j \times I \to N \times \{\delta_i\} \times I$ which satisfy the hypotheses of 4.2. We conclude therefore that $X \subset M$ is locally flat at x.

4.4 LEMMA. *If $X \subset M$ satisfies the hypotheses of* 3.4.1, *then $X \times \mathbf{R}^2 \subset M \times \mathbf{R}^2$ is locally flat.*

Proof. Since X is a manifold factor, $X \times \mathbf{R}^2$ is a topological manifold. Restrict attention to $U \subset X \times \mathbf{R}^2$ which is either an open disc in the interior, or $\mathbf{R}^j \times (0, 1]$ which intersects $\partial X \times \mathbf{R}^2$ in $\mathbf{R}^j \times \{1\}$. Let W be an open set in $M \times \mathbf{R}^2$ so that $W \cap X \times \mathbf{R}^2 = U$, and W retracts to U by $r\colon W \to U$. Then $r\colon W - U \to U$ has an end which is 0-LC and either 1-LC (codimension $\geqq 3$) or has constant fundamental group $\mathbf{Z}$ (codimension 2). This end is tame if the codimension $\geqq 3$ by the homological argument of the proof of 3.1.1 using the fact that M is a homology manifold. In codimension 2 this follows directly from local homotopy unknottedness.

Local homotopy unknottedness can be described as follows: duality implies that there is a regular $\mathbf{Z}$ cover of the end of $M - X$. This cover is assumed to be k-LC (the straightforward extension of 1-LC) for every k. We note that this condition can be weakened to 1-LC and homologically k-LC via the eventual Hurewicz theorem of Section 5. A simple homology argument shows from this that the end is homologically tame, hence tame.

We conclude from the end theorem that U has a mapping cylinder neighborhood on W, which extends the product neighborhood $\partial U \times R^{j+1} \subset \partial W$ if U has boundary. Let $f: N \to U$ be the map. f is an approximate S^j fibration. This follows from the fact that if $V \subset U$ is any open set, $(V_f, f^{-1}(V))$ and V both satisfy Poincaré duality, so the homotopy fiber of $f^{-1}(V) \to V$ is a sphere (Browder [1], Chapter I, § 4).

Since we have arranged U to be a PL manifold, it follows from 3.3.2 variation 4 that there are concordances $h_i: N \times I \to U$ rel ∂N of f to an S^j block bundle, of diameter $<1/i$. Since U is contractible, a block bundle is trivial so we can assume it is a product $U \times S^j$.

This gives isomorphisms $g_i: U \times S^j \simeq N$ so that $d(p, fg_i) < 1/i$. We fill in between these using the relative form of 3.3.2. The juxtaposition of the concordances $(h_i)(-h_{i+1})$ gives an approximate fibration concordance between g_i and g_{i+1}. Use 3.3.2 to approximate it by a block bundle over $U \times I$. The trivializations of this bundle over $U \times \{0,1\}$ extend to $U \times I$, again because U is contractible. This yields an isomorphism $G_i: U \times S^j \times I \to N \times I$ with

$$G_i | U \times S^j \times \{0\} = g_{i+1}\,, \qquad G_i | U \times S^j \times \{1\} = g_i\,.$$

Clearly we can choose the concordances small enough that $d(p, fG_i) < 1/i$.

These maps G_i satisfy the hypotheses of Lemma 4.2, so $U \subset W$ is locally flat. It follows that $X \times \mathbf{R}^2 \subset M \times \mathbf{R}^2$ is locally flat, as required for the lemma.

Lemmas 4.3 and 4.4 combine in the obvious way to prove 3.4.1.

5. Homotopy theory

The principal result of this section is the "eventual Hurewicz theorem" (over X). This is used to give homological characterizations of tameness of an end (1.1), a tame structure (2.4), and thin h-cobordisms (2.6).

It is well known that a map which is zero on homology may not be contractible. However this is eventually true in the sense that if we have a long enough sequence $A_0 \to A_1 \to \cdots \to A_k$, each of which is zero on homology, then the composition is contractible. This result is due to S. Ferry [F1, Prop. 3.1]. The theorem here is a relative form, with δ control over a space X.

5.1 *Definition.* Suppose X is a metric space, $C \subseteqq X$, and there is a commutative diagram

$$\begin{array}{ccc} (A_0, B_0) & \xrightarrow{f} & (A_1, B_1) \\ & {\scriptstyle e_0}\searrow \quad \swarrow{\scriptstyle e_1} & \\ & X & \end{array}$$

:

(a) f is *relatively* (δ, k) *connected over* C if for every map of a relative k complex $(R, S) \xrightarrow{r} (A_0, B_0)$ with $e_0 r(R) \subseteq C$, fr is homotopic rel S into B_1, by a homotopy of diameter $< \delta$ in X.

(b) f is *relatively homologically* (δ, k) *trivial over* C if for every $K \subseteq C$ the homomorphism

$$f_*: H_j\left(e_0^{-1}(K), e_0^{-1}(K) \cap B_0\right) \longrightarrow H_j\left(e_1^{-1}(K^\delta), e_0^{-1}(K^\delta) \cap B_1\right)$$

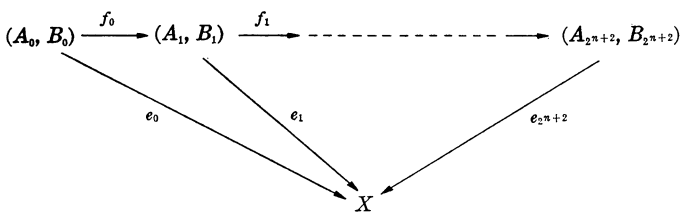
is zero for $j \leqq k$.

Here H_j denotes singular homology with $\mathbf{Z}$ coefficients.

Recall from Section 2 that a map over X is $(\delta, 1)$ connected over $C \subseteq X$ if it has the δ approximate lifting property for relative 2-complexes in C. In particular notice the distinction between $(\delta, 1)$ connected and relative $(\delta, 1)$ connectedness.

5.2 THEOREM (*eventual Hurewicz theorem*). *Suppose X is a compact metric space, $X \supseteq C$, $\varepsilon > 0$, n is given, and X is locally 1-connected in a neighborhood of C^ε. Then there is $\delta > 0$ such that the following is true:*

Suppose there is a diagram



and $A_{2^{n+2}}$ has a regular covering space with group π such that

1) *each $A_i^\wedge \to A_{i+1}^\wedge$ and $B_i^\wedge \to B_{i+1}^\wedge$ is $(\delta, 1)$ connected over C^ε, and*

2) *each $f_i^\wedge$ is relatively (δ, n) homologically trivial over C^ε.*

Then $(A_0, B_0) \to (A_{2^{n+2}}, B_{2^{n+2}})$ is relatively (ε, n) connected over C.

Proof. We actually prove that 2 compositions are needed if $n = 0$, 4 if $n = 1$, and $2^{n+2} - 2$ if $n \geqq 2$. The proof proceeds by induction on n after the first two steps.

$n \leqq 1$. First note that since X is locally 1-connected given $\gamma > 0$, there is a δ so that if $r_1, r_2: (R, S) \to X$ are maps of a relative 1-complex which agree on S and $d(r_1, r_2) < \delta$, then they are homotopic rel S by a homotopy of diameter $< \gamma$.

Suppose $(R, S) \to (A_0, B_0)$ is a map, and denote the induced covers by $r^\wedge: (R^\wedge, S^\wedge) \to (A_0^\wedge, B_0^\wedge)$. There is a diagram

$$\begin{array}{ccc} S\hat{} & \xrightarrow{r} & B_0\hat{} \\ \cap & & \downarrow e_0\hat{} \\ R_0\hat{} & \xrightarrow{e_0\hat{}r} & X\,. \end{array}$$

Since $B_0\hat{} \to B_1\hat{}$ is $(\delta, 1)$ connected and π acts freely on $R\hat{}$, we can find a π equivariant S lift of the 0 skeletion of $R - S$. (Choose a single lift for each 0-simplex in $R - S$ and get lifts for others in $(R - S)\hat{}$ by translating by π). Call this lift $s\hat{}: R_{(0)}\hat{} \to B_1\hat{}$.

By choice of δ, since $d(e_1 s, e_0 r) < \delta$, they are γ homotopic in X rel S. The homotopy defines a diagram

$$\begin{array}{ccc} R_{(0)}\hat{} \times \{0, 1\} \cup S\hat{} \times I & \longrightarrow & A_1\hat{} \\ \cap & & \downarrow \\ (R_{(0)} \cup S)\hat{} \times I & \longrightarrow & X\,. \end{array}$$

Since the left side is a relative 1-complex, there is a π equivariant δ lift into $A_2\hat{}$.

Dividing by π we get a $\gamma + \delta$ homotopy $(R_{(0)} \cup S) \times I \to A_2$ which takes $f_1 f_0 r$ into B_2 rel S. If we choose δ, γ small enough that $\gamma + \delta < \varepsilon$, this is the δ required for the case $n = 0$.

For $n = 1$ one just repeats this argument for the 1-cell of (R, S). Two more compositions are required.

$n \geqq 2$. We suppose the theorem is true for $n - 1$, and that $(R, S) \to (A_0, B_0)$ is a relative n complex over C. Subdivide so that the n-cells have diameters less than some $\gamma < \varepsilon/2$ (and satisfying other properties identified as we go along). Suppose δ is small enough that after t compositions we can deform the $n - 1$ skeleton into B_t by a homotopy of diameter $< \gamma$. This leaves the n cells to be pulled in. Let $(D^n, S^{n-1}) \to (A_t, B_t)$ be one of these. This is a map of diameter $< 3\gamma$ in X, and has image in C^r.

Lift this map to $(D^n, S^{n-1}) \to (A_t\hat{}, B_t\hat{})$. Then the homological triviality hypothesis implies that there is a factorization $(D^n, S^{n-1}) \to (L, M) \to (A_{t+1}\hat{}, B_{t+1}\hat{})$ such that (D^u, S^{u-1}) is homologically trivial in (L, M), and the image of L in X has diameter $< 3\gamma + 2\delta$.

Now presume that γ, δ are small enough that a map of a 1-complex into X of diameter $< 3\gamma + 2\delta$ is nullhomotopic by a map of diameter $< \rho$ (ρ to have conditions imposed as we proceed). Again we are using the local 1-connectedness of X.

This presumption gives a map of $L' = L \cup c(L_{(1)})$ into X of diameter $< 2\delta + 3\gamma + \rho$ and a diagram

$$\begin{array}{ccc} L & \longrightarrow & \hat{A}_{t+1} \\ \cap & & \downarrow \\ L' & \longrightarrow & X\,. \end{array}$$

Since $\hat{A}_{t+1} \to \hat{A}_{t+2}$ is $(\delta, 1)$ connected, this lifts to $(L', M) \to (\hat{A}_{t+2}, \hat{B}_{t+2})$ of diameter $4\delta + 3\gamma + \rho$.

Next we suppose δ, γ are small enough so that we can choose ρ small enough so that we can apply the induction hypothesis to conclude that the $n-1$ skeleton of (L', M) deforms into $\hat{B}_{2t+2}$ inside $\hat{A}_{2t+2}$, by a deformation of diameter $<\varepsilon/2$. This gives a factorization

$$(D^n, S^{n-1}) \longrightarrow \big(L', M \cup L'(n-1)\big) \longrightarrow (\hat{A}_{2t+2}, \hat{B}_{2t+2})\,.$$

However if $n \geqq 3$, $\big(L', M \cup L'(n-1)\big)$ is a pair of 1-connected complexes which is relatively $n-1$ connected. The ordinary Hurewicz theorem implies that

$$\pi_n\big(L', M \cup L'(n-1)\big) = H_n\big(L', M \cup L'(n-1)\big)$$

so since (D^n, S^{n-1}) is trivial in homology, it is homotopic into $M \cup L'(n-1)$. Since this maps into $\hat{B}_{2t+2}$, this gives a deformation of the n cell into $\hat{B}_{2t+2}$.

Dividing by π we get a deformation into B_{2t+2}, of diameter $4\delta + 3\gamma + P + \varepsilon/2$. If these add up to less than ε, since $2t+2 = 2(2^{n+1}-2)+2 = 2^{n+2}-2$, the induction step is complete.

The argument for $n=2$ is not quite finished, since we assumed $n \geqq 3$ so that $M \cup L'(n-1)$ would be 1-connected. In $n=2$ we can make this 1-connected by adding the cone on the 1 skeleton as in the case $n=1$ above. This requires another composition, so $n=2$ requires $2t+3$. However as indicated above $t=4$ works for $n=1$, so $2(4)+3 = 2^{2+2}-5 < 2^{2+2}-2$, and the induction starts off properly anyway.

This completes the proof of 5.2. We note that the number of compositions used can be reduced considerably.

We now apply this theorem to obtain homological characterizations of the tameness hypothesis of Section 2.

First weaken the definition of a tame structure for an approximate end (Definition 2.4) to a "homological tame structure" by replacing condition (3) with

(3′) For every $K \subset C$ and $0 \leqq j \leqq k-2$,

$$H_*\big(e^{-1}(K) - W_j,\, e^{-1}(K) - U_{j+1};\, \mathbf{Z}\pi\big)$$
$$\longrightarrow H_*\big(e^{-1}(K^\delta) - W_{j+1},\, e^{-1}(K^\delta) - U_{j+2},\, \mathbf{Z}\pi\big) \text{ is zero.}$$

5.3 PROPOSITION. *Suppose X is a compact metric space, $X \supseteq C$, $\varepsilon > 0$, n is given, and X is locally 1-connected in a neighborhood of C^ε. Then there*

is $\delta > 0$ such that if $e: M^n \to X$ is a map and $\{U_i, W_i\}$ is a homological $(\delta, 2^{n+2}k)$ tame structure for e over C^ε, then the subsequence of terms with index divisible by 2^{n+2} is an (ε, k) tame structure over C.

Proof. This follows easily by applying 5.2 to the compositions

$$(U_j - W_j,\ U_j - W_{j+2^{n+2}}) \longrightarrow (U_j - W_{j+1},\ U_j - U_{j+2^{n+2}+1}) \longrightarrow \cdots \longrightarrow (U_j - W_{j+2^{n+2}},\ U_j - U_{j+2(2^{n+2})}) .$$

The definition of a (δ, h)—cobordism can similarly be weakened. Suppose $e: (M, \partial_0 M) \to X$, and $M\hat{} \to M$ is a regular π cover. We say e is a $(\delta, H_*\mathbf{Z}\pi)$-cobordism over $C \subseteq X$ if for every $K \subseteq C$

$$H_*\big(e^{-1}(K),\ e^{-1}(K) \cap \partial_0 W;\ \mathbf{Z}\pi\big) \longrightarrow H_*\big(e^{-1}(K^\delta),\ e^{-1}(K^\delta) \cap \partial_0 W;\ \mathbf{Z}\pi\big) \text{ is zero.}$$

5.4 PROPOSITION. *Suppose X is a compact metric space, $X \supseteq C$, $\varepsilon > 0$, n is given, and X is locally 1-connected in a neighborhood of C^ε. Then there is $\delta > 0$ so that if $e: (M^n, \partial_0 M) \to X$ is proper, $M\hat{} \to M$ is a regular π cover of M with $M\hat{}$, $\partial_0 M\hat{}$, $\partial_1 M\hat{}$ all $(\delta, 1)$ connected over C^ε, and $(M, \partial_0 M)$ a $(\delta, H_*\mathbf{Z}\pi)$-cobordism over C^ε, then $(M, \partial_0 M)$ is an (ε, h)-cobordism over C^ε.*

Proof. Suppose $\gamma > 0$. Then there is a δ such that 5.2 applied to the 2^{n+2}-fold composition of the identity map $(M, \partial_0 M) \to (M, \partial_0 M)$ with itself yields a γ deformation of $e^{-1}(C^{\varepsilon-\gamma})$ into $\partial_0 M$.

The next step is to use duality to show $M, \partial_0 M$ is also an $H_*\mathbf{Z}\pi$-cobordism. Suppose $K \subset C^{\varepsilon-3\delta}$. Then the deformation above keeps $e^{-1}(C^{\varepsilon-\gamma} - K^{2\gamma})$ in $e^{-1}(C^\varepsilon - K^\gamma)$. This shows that

$$\big(M,\ \partial_0 M \cup e^{-1}(X - K^{2\gamma})\big) \longrightarrow \big(M,\ \partial_0 M \cup e^{-1}(X - K^\gamma)\big)$$

is homotopically trivial and so in particular is zero on $H^*(;\mathbf{Z}\pi)$. The cohomology of this map is Alexander-Poincaré dual to

$$H_*\big(e^{-1}(K^\gamma),\ e^{-1}(K^\gamma) \cap \partial_1 M;\ \mathbf{Z}\pi\big) \longrightarrow H_*\big(e^{-1}(K^{2\gamma}),\ e^{-1}(K^{2\gamma}) \cap \partial_1 M;\ \mathbf{Z}\pi\big) .$$

We therefore conclude that $(M, \partial_1 M)$ is a $(2\gamma, H_*\mathbf{Z}\pi)$ cobordism over $C^{\varepsilon-3\gamma}$. If γ is small enough, the argument above shows there is an ε retraction of M to $\partial_1 M$ over C, so M is an (ε, h) cobordism.

Finally, homological tameness of an end has been defined in Definition 1.6. There is an evident relative version.

5.5 PROPOSITION. *Suppose X is compact metric, $X \supseteq C$, and X is locally 1-connected in a neighborhood of C. If the end of $e: M \to X$ is homologically tame over a neighborhood of C, then for any neighborhood U of the end of e, any k, and sufficiently small $\delta > 0$, there is a $(\delta, k, \varnothing)$ tame structure over C^δ, contained in U. In particular the end is tame over a neighborhood of C.*

Proof. In view of 5.4, it will be sufficient to produce a homological tame structure. The major part of the construction is involved with the $(\delta, 1)$ connectedness hypothesis.

Suppose we know the following (where $U^\wedge$ denotes the cover induced from $M^\wedge$ of the definition):

1) Given U, there is $V \subset U$ such that $V^\wedge \to U^\wedge$ is $(\delta, 1)$ connected over C^δ.

2) Given $U_0 \supset W_0 \supset W_1 \supset W_2 \supset U_1 \supset U_2$ with all inclusions $(\delta/3, 1)$ connected over $C^{2\delta}$, then there is $V \subset U_2$ so that $U_1^\wedge - U_2^\wedge \to U_0^\wedge - V^\wedge$ is $(\delta, 1)$ connected over C^δ.

Given this, we can construct a homological tame structure by induction. Suppose we are given

$$U_0 \supset U_1 \supset \cdots \supset U_{n-2} \supset W_0 \supset W_1 \supset W_2 \supset U_{n-1} \supset V_0 \supset V_1 \supset V_2 \supset U_n$$

so that the U's form a tame structure and all inclusions are $(\delta, 1)$ connected. Use (1) to find $U_n \supset T_0 \supset T_1 \supset T_2 \supset T_3$. Use $U_{n-2} \supset W_0 \supset W_1 \supset W_2 \supset U_{n-1} \supset U_n$ to find $R_0 \subset U_n$. Use the homological tameness to find $S_0 \subset U_n$. Finally let U_{n+1} be a neighborhood contained in $T_3 \cap S_0 \cap R_0$; then $U_0 \supset \cdots \supset U_{n-1} \supset V_0 \supset V_1 \supset V_2 \supset U_n \supset T_0 \supset T_1 \supset T_2 \supset U_{n+1}$ satisfy the induction hypothesis.

Proof of (1). Suppose δ is so small that $M^\wedge \to X$ is onto and 1-LC over $C^{2\delta}$. Choose $V_1 \subset U$ and δ_1 so that loops in $V_1^\wedge \cap e^{-1}(C^{2\delta - \delta_1})$ of diameter $< \delta_1$ in X extend to discs in $U^\wedge$ of diameter $< \delta/2$. Choose $V_2 \subset V_1$ and δ_2 so that points in $V_2^\wedge \cap e^{-1}(C^{2\delta - \delta_1 - \delta_2})$ closer than δ_2 in X can be joined by arcs of diameter $< (1/2)\delta_1$ in $V_1^\wedge$. Then $V_2^\wedge \to U^\wedge$ is $(\delta, 1)$ connected over C^δ, as can be seen by subdividing a relative 2-complex and lifting the skeletons.

Proof of (2). We construct a "universal" 2-complex which is finite, lift it, and use compactness.

Part of the hypothesis is that the group π of the cover $M^\wedge$ is finitely presented. Let $g_1, \cdots, g_k$ be generators and $R_1, \cdots, R_j$ be relations for π.

Suppose V_1 is a neighborhood of the end whose closure is contained in U_3. Since $M - V_1$ is compact, there is a finite n-complex K, $U_1 - U_2 \subset K \subset W_2 - V_1$. Suppose $\gamma > 0$ and subdivide K so that the simplices have diameter $< \gamma$. Let S be the 1-skeleton of K over $C^{2\delta}$. S has a regular π cover $S^\wedge \to S$ induced from $M^\wedge$.

Choose a finite set of points in C^δ so that each point of C^δ is within γ of one of these points or the image of a vertex of S. Define R_0 to be union of S with the set, and $R_0^\wedge$ the evident π cover extending $S^\wedge$.

Next suppose $\alpha > 0$, and that γ is small enough that points within 3γ

in $C^{2\delta}$ can be joined by an arc of diameter $<\alpha$. We construct R_1 by adding two types of arcs.

(a) Add one arc between each pair of distinct vertices of R_0 closer than 3γ in X.

(b) At each vertex v, add k loops $l_{v,1}, \cdots, l_{v,k}$ corresponding to the generators of π.

Arcs of type (a) are mapped into X using the hypothesis; type (b) are projected to the image of the vertex. A π cover $R_1^\wedge \to R_1$ is constructed by lifting an arc of type (a) to an arc between arbitrary points in $R_0^\wedge$ over the endpoints, and adding π translates. An arc $l_{v,j}$ of type (b) lifts to an arc from some v' lying over v, to $g_j v'$, and then π translates of this arc.

Finally suppose α is small enough that loops in $C^{2\delta}$ of diameter $<3\alpha$ extend to discs of diameter $<\delta/6$. We construct R_2 by adding 3 types of 2-discs to R_1.

(a) Suppose h is a 1-cell of R_1 with distinct endpoints. Then for each j we add a 2-disc on the loop $(h)(l_{h(1),j})(h^{-1})(l^{-1}_{h(0),j})$.

(b) At each vertex v we add 2-cells to compositions of the $l_{v,*}$ corresponding to the relations given for π.

(c) Suppose that l is an embedded loop in R_1, with ends at a vertex v, and of diameter $<3\alpha$. Lift this into $R_1^\wedge$, then the ends differ by an element $P \in \pi$. Choose (arbitrarily) some word representing P, and then add a 2-cell on l composed with the inverse of this word in the loops $l_{v,*}$.

This produces a finite 2-complex R_2. We map R_2 to X and construct a regular π cover $R_2^\wedge \to R_2$ in the evident way.

We claim that $(R_2^\wedge, S^\wedge)$ is $(\delta/3, 1)$ connected. By now the reader should be able to verify this, so we omit it.

Returning to the proof of (2), use the $(\delta/3, 1)$ connectedness $U_0 \supset W_0 \supset W_1$ to lift

$$\begin{array}{ccc} S^\wedge & \longrightarrow & U_0 \\ \cap & & \downarrow \\ R_2^\wedge & \longrightarrow & C^\delta \subset X \end{array}$$

π equivariantly (as before lift representatives, use π action to lift the rest). This gives a $2\delta/3$ lift $R_2 \to U_0$. Since R_2 is finite the image is contained in the complement of some neighborhood V of the end. Now the inclusion $U_1^\wedge - U_2^\wedge \to U_0^\wedge - V^\wedge$ is $(\delta, 1)$ connected, since in fact a relative 2-complex lifts into $R_2^\wedge \to U_0^\wedge - V^\wedge$.

This completes the proof of 5.5 (and therefore 1.7). As a final remark we note that the group π appearing in a tame structure is automatically

finitely presentable:

5.6 PROPOSITION. *Suppose* $A_0 \xrightarrow{f_0} A_1 \xrightarrow{f_1} A_2$ *are connected finite complexes, and* $\operatorname{im}\pi_1(f_0) \to \operatorname{im}\pi_1(f_1)$ *is an isomorphism. Then* $\operatorname{im}\pi_1(f_0)$ *is finitely presented.*

Proof. Composition of $\pi_1 A_1 \xrightarrow{\pi_1 f_1} \pi_1 A_2$ with the inverse of the isomorphism given yields a projection on $\pi_1 A_1$. It follows that the image and kernel are complementary direct factors. But a direct factor of a finitely presented group is finitely presented.

6. Geometrical connectivity

In this section we show that if $(N, \partial_0 N) \to X$ is relatively (δ, k) connected, then it has a handlebody structure of diameter $<\varepsilon$ with no handles of dimension $\leqq k$. This and a theorem from Section 8 are used to prove the thin h-cobordism theorem. The geometrical connectivity theorem will also be a main ingredient in the proof of the approximate end theorem in the next section.

This is an ε, δ version of a standard theorem. The most lucid account of the classical result is due to C.T.C. Wall [1], and we will follow his development fairly closely.

A handlebody structure on a manifold $(N, \partial_0 N)$, with $\partial_0 N$ a closed codimension 0 submanifold of ∂N, is a filtration $N_{-1} \subset N_0 \subset \cdots \subset N_n$ where n is the dimension of N, such that N_{-1} is a collar of $\partial_0 N$, and for $i \geqq 0$, N_i is obtained from N_{i-1} by attaching i handles. As a further notation we define $\partial_0 N_i = \partial_0 N$, and $\partial_1 N_i =$ closure of $\partial N_i - \partial_0 N_i$. Note that the $i+1$ handles are attached to $\partial_1 N_i$.

If $e\colon N \to X$ is a map to a metric space, we say the handlebody structure has diameter $<\delta$ in X if the collar N_{-1} has diameter $<\delta$ when thought of as a homotopy of $\partial_0 N$, and each handle (component of $N_i - N_{i-1}$) has image diameter $<\delta$.

6.1 THEOREM (*Geometrical connectivity*). *Suppose X is a compact metric space, $X \supset C \supset D$, $\varepsilon > 0$, and $n \geqq 5$. Then there is $\delta > 0$ such that if $-1 \leqq k < n-3$ and $(N, \partial_0 N) \to X$ is a compact n manifold which is relatively (δ, k) connected over C^ε (Definition 5.1), and has a handlebody structure which over D^ε has diameter $<\delta$ and no handles of dimension $\leqq k$, then there is a handlebody structure of diameter $<\varepsilon$ which has no handles of dimension $\leqq k$ over C, and which agrees with the one given over D.*

As usual this theorem applies equally in the differentiable, PL, or topological categories.

6.2 COROLLARY. *If in addition to the conditions above* $(N, \partial_1 N)$ *is relatively* (δ, j) *connected over* C^ϵ, *then there is a handlebody structure with handles on* C *only in dimensions* i, $k < i < n - j$, *provided* $j < n - 3$ *and* $k + j < n - 2$.

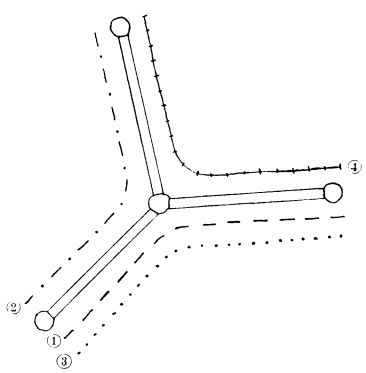
The hypothesis $j < n - 3$ is symmetric with the hypothesis on k. The hypothesis $k + j < n - 2$ implies that there are at least two adjacent dimensions in which handles may occur.

Proof of 6.1. The proof is by induction on k after the first two steps.

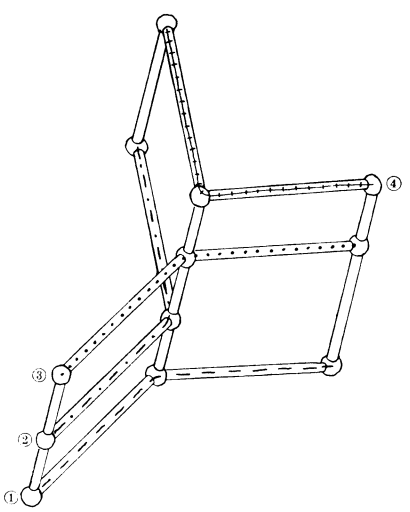
$k = -1$. A handlebody structure with small handles and collar may be obtained from an arbitrary one by subdividing and some small isotopies (to ensure that handles are attached to lower dimensional ones). We omit the details.

$k = 0$. We must eliminate the 0-handles. If $(N, \partial_0 N)$ is $(\delta, 0)$ connected over C^ϵ, then there is a path of diameter $<\delta$ from each 0-handle to the collar. We can push these into the 1 skeleton with diameter $<n\delta$, so each 0-handle can be joined to the collar by a sequence of 0 and 1-handles of total diameter $<(n + 2)\delta$. We can presume there are no repetitions of 0-handles in a sequence by discarding loops. Also discard the sequence after the first intersection with the collar.

The next step is to subdivide the 1 skeleton to make all these sequences disjointly embedded. Order the 0-handles; then the sequences are ordered by the 0-handle they terminate in. If a 0-handle appears in m sequences, replace it by m 0-handles and $m - 1$ 1-handles: $D^{n-1} \times I_m$, where I_m is the unit interval with m 0-handles. These 0-handles are ordered, and we



A 0-handle occurring in 4 sequences



The subdivided handles

associate them with the sequences in the order preserving way. Next if a 1-handle appears in p sequences, replace it by p 1-handles and $p-1$ 2-handles: replace $D^{n-1}\times I$ by $(D^{n-2}\times I_p)\times I$. Again the 1-handles are ordered and we associate them with the sequence in the order preserving way.

Now we claim that after an isotopy of these identifications, the new 1-handles associated to a sequence are attached to either the collar or 0-handles associated with the same sequence. This is because the orderings agree.

These new handles all have diameter $<\delta$. The result is not quite a handlebody because some of the old 2-handles may be added onto the new ones. They can be isotoped off by a small isotopy, which increases the size to less than 3δ. Now each 0-handle occurs in a unique sequence of 0 and 1-handles. The union of a sequence is an n disc, intersecting the collar in an $n-1$ disc. Unions of distinct sequence are disjoint, so their union with the collar is again a collar, with diameter $<(n+4)\delta$.

If we choose $\delta=(1/n+4)\varepsilon$, this is the desired handlebody structure without 0-handles.

$k>0$. Assume (by induction) that $(N, \partial_0 N)$ is relatively (δ, k) connected over C^ε, and has a handle decomposition of diameter $<\delta$ with no handles of dimension $\leqq k-1$ over C^ε. We will eliminate the k handles by introducing cancelling pairs of $k+1$, $k+2$ handles, and then using the new $k+1$ handles to cancel the k handles over C.

Over C^ε there are no handles $\leqq k-1$, so the components of the closure of $N_k - N_{-1}$ over C^ε are k handles. Denote these by $D^k_\alpha\times D^{n-k}_\alpha$, for α in some index set. Define $M=\partial_1 N_k\cap\partial_1 N_{k+1}$, then M is also $\partial_1 N_{k+1}-\bigcup\operatorname{int}(D^k_\alpha\times S^{n-k-1}_\alpha)$ so choosing a point $P_\alpha\in S^{n-k-1}_\alpha$ gives inclusions $(D^k_\alpha, S^{k-1}_\alpha)\to(\partial_1 N_{k+1}, M)$, of diameter $<\delta$. Our goal for a while is to find small relative nullhomotopies of these in $(\partial_1 N_k\cup\{k+1 \text{ handles}\}, M)$.

First since $M\subset\partial_1 N_{-1}$ (there are no handles $\leqq k-1$) we can push the boundaries of these inclusions down through the collar to get $(D^k_\alpha, S^{k-1}_\alpha)\to(N, M^1)$. Here M^1 is the copy of M at the bottom of the collar. Since $(N, \partial_0 N)$ is relatively (δ, k) connected, there is a homotopy of this map rel S^{k-1}_α into $\partial_0 N$ of diameter $<\delta$. This gives a map $\rho_\alpha\colon(D^{k+1}, S^k_+, S^{k-1})\to(N, \partial_0 N, M^1)$ of diameter $<5\delta$. Now push ρ_α back up through the collar, and into the $k+1$ skeleton. This gives $\rho_\alpha\colon(D^{k+1}; S^k_+, S^{k-1})\to(\partial_1 N_{-1}\cup\{k \,\&\, k+1 \text{ handles}\}, \partial_1 N_{-1}, M)$ of diameter $<(7+n-k)\delta$, which is the original inclusion on S^{k-1}_-.

In the special case that $(k+1)+k<n$, we can make ρ_α disjoint from

the k handles by transversality, and push into $(\partial_1 N_k \cup \{k+1 \text{ handles}\}, M)$ to get the desired nullhomotopy. In particular if $n \geqq 6 > (2+1)+2$ this works for 1 and 2 handles, and if $n = 5$ it works for 1-handles.

Suppose then that $k > 2$. We can make ρ_α disjoint from the center points of the k handles and push out to the boundary to get

$$(D^{k+1}, S^k_+, S^{k-1}) \longrightarrow (\partial_1 N_k \cup \{k+1 \text{ handles}\} \cup \bigcup_\alpha S^{k-1}_\alpha \times D^{n-k}_\alpha, M \cup \bigcup_\alpha S^{k-1}_\alpha \times D^{n-k}_\alpha, M) .$$

This is related to what we want by an excision, so we pass to homology to use that fact.

Let $K = e(\rho_\alpha(D^{k+1}))^\delta$. Then if we let A be the union of the $S^{k-1}_\beta \times D^{n-k}_\beta$ which are contained in $e^{-1}(K)$, and let B be the union of the $k+1$ handles contained in $e^{-1}(K)$, we see that ρ_α has image in

$$\big((\partial_1 N_k \cap e^{-1}(K)) \cup B \cup A, M \cap e^{-1}(K) \cup A, M \cap e^{-1}(K)\big) .$$

It therefore defines a nullhomotopy of the composition

$$(D^k_\alpha, S^{k-1}_\alpha) \longrightarrow \big((\partial_1 N_k \cap e^{-1}(K)) \cup B, M \cap e^{-1}(K)\big) \longrightarrow \big((\partial_1 N_k \cap e^{-1}(K)) \cup B \cup A, (M \cap e^{-1}(K)) \cup A\big) .$$

Since $2 < k < n-3$, all these spaces in the last two pairs have the same fundamental group, say π. Moreover the pairs are $k-1$ connected. Finally since it is an excision with $\mathbf{Z}\pi$ coefficients, it induces an isomorphism on the relative π_k groups. Therefore we conclude that $(D^k_\alpha, S^{k-1}_\alpha) \to (\partial_1 N_k \cup \{k+1 \text{ handles}\}, M)$ is nullhomotopic over k. This gives a map with diameter $<(11+n-k)\delta$.

This achieves the goal set above. The next goal is to use these homotopies to prove: There are (small $k+1$) discs B_α disjointly embedded in $\partial_1 N_{k+1}$, with ∂B_α in M, and there is an isotopy of $\partial_1 N_k$ which carries each ∂B_α to a sphere which intersects $\{0\} \times S^{n-k-1}_\alpha$ in exactly one point (here $0 \in D^k_\alpha$), and is disjoint from the other S^{n-k-1}. Once we have this we will almost be finished.

The first step is to make the homotopies ρ_α found above disjoint from the center point of the $k+1$ handles, and push out to the boundary. This increases diameter by $<2\delta$, and gives maps $\rho_\alpha : D^{k+1} \to \partial_1 N_k \cup \partial_1 N_{k+1}$, with $\rho_\alpha | S^k_- = D^k_\alpha$, and $\rho_\alpha(S^k_+) \subset M$.

Since $\partial_1 N_k \cup \partial_1 N_{k+1}$ is $\partial_1 N_k$ union with many small copies of $D^{k+1} \times S^{n-k-2}$, we can assume the ρ_α intersect the components of $\partial_1 N_{k+1} - \partial_1 N_k$ in disjointly embedded $(k+1)$ discs. Join the preimages of these discs in D^{k+1} by arcs. Then since $n-k > 2$ we can deform these arcs rel ends into M. Since M is an $n-1$ manifold, $n-1 \geqq 5$, we can approximate ρ_α on a neighborhood of

these arcs by disjoint embeddings. This makes ρ_α an embedding on a disc into $\partial_1 N_{k+1}$, and the remainder of the image of ρ_α is in $\partial_1 N_k$. Call this disc B_α.

We have part of our goal: disjoint discs in $\partial_1 N_{k+1}$. The maps ρ_α on the complement of the disc in D^{k+1} define homotopies from ∂B_α in $\partial_1 N_k$ to $\rho_\alpha | S^k$. By construction $\rho_\alpha | S^k$ intersects $\{0\} \times S_\alpha^{n-k-1}$ in exactly one point and is disjoint from the other S_β^{n-k-1}. Since $k < n-3$ we can appeal to the Stallings lemma [1], or the Whitney trick to get an isotopy of $\partial_1 N_k$ which carries ∂B_α to a sphere with the same intersections. We point out that the proofs of these isotopy theorems construct isotopies which are constant except in a neighborhood of a 2-complex contained in the given homotopy. Therefore the isotopy constructed has diameter only a little larger than the homotopies, so the discs and isotopy have diameter $<(15 + n - k)\delta$.

The proof can now be completed. Use the $k+1$ discs $B_\alpha \subset \partial_1 N_{k+1}$ to add cancelling pairs of $k+1$ and $k+2$ handles. This gives a handle structure on N of diameter $<(17 + n - k)\delta$. (Only the new handles are the large, however.) Apply the isotopy in $\partial_1 N$. This may increase the diameter of the handles to $<3(17 + n - k)\delta$. Now, however, each k handle is paired to a $k+1$ handle which intersects it in a transverse disc. The union of such a pair is a disc intersecting the collar in a disc. These unions are all disjoint, so they may be absorbed into the collar (thereby increasing its diameter). This provides a handlebody with no k handles over C^γ, $\gamma = \varepsilon - 3(18 + n - k)\delta$. Therefore we conclude that $\delta = (1/3(18 + n - k))\varepsilon$ satisfies the conclusion of the induction step.

This completes the proof of 6.1. Notice that we get an explicit formula for δ from the proof ($\delta = (1/3(18 + n))^k \varepsilon$ works). This is because in contrast to proofs in Sections 5, 7, 8, we have not used local 1-connectedness in X.

Proof of 6.2. We note that 6.2 follows from 6.1 applied to both $(N, \partial_0 N)$ and $(N, \partial_1 N)$ (after some fiddling with collars), and the observation that in the proof the process of eliminating handles of dimension $\leqq k$ does not change the handles of dimension $> k+2$.

6.3 *Proof of the thin h-cobordism theorem.* Suppose $e: (N, \partial_0 N) \to X$ is a (δ, h) cobordism over C^ε. Then both $(N, \partial_0 N)$ and $(N, \partial_1 N)$ are relatively (δ, k) connected for all k over C^ε, so we may apply 6.2 to eliminate all but 2 layers of handles over C.

Suppose then, that $(N, \partial_0 N)$ is a δ h-cobordism over C^ε, and it has a handlebody decomposition of diameter $<\delta$ which is a product over D^ε and has only k, $k+1$ handles over C^ε. We will get rid of these last handles by

translating the problem into "algebra" and applying the theorem of Section 8.

Let $N^\wedge \to N$ be the $(\delta, 1)$ connected cover with group π. Let G_k be the free $\mathbf{Z}\pi$ module generated by a set of points in X, one in the image of each k handle in C^ε, and define G_{n+1} using the $k+1$ handles. These are *geometric modules* in the terminology of Section 8.

The boundary homomorphism $\partial: G_{n+1} \to G_n$ in the chain complex associated to the handlebody structure is defined by choosing an oriented lift in $N^\wedge$ of each handle, and looking at algebraic intersections of translates by elements of π. Since handles do not intersect algebraically if they are disjoint geometrically, and since the diameter of the handle decomposition is $<\delta$, we see that δ has diameter $<3\delta$ (see Definition 8.1 for the *diameter* of ∂).

Next consider a homotopy $e^{-1}(C^\varepsilon) \times I \to N$, which is a δ homotopy rel $\partial_0 N$. As usual a homotopy defines a chain homotopy on the chain complex. In this case we get only $g: G_n \to G_{k+1}$ so that ∂g restricted to $G_k | C^{\varepsilon - 5\delta}$ is the inclusion. Since the chain homotopy can also be defined by geometric intersections, g has diameter $<5\delta$. This shows that ∂ is a 6δ isomorphism over $C^{\varepsilon/2}$. We can now apply the geometric group Theorem 8.4. This tells us that if $\varepsilon/2 > \gamma > 0$, then there is a δ so small that there is a geometric module G' on $X - D^{\varepsilon/2}$ and a deformation H of $G \oplus G'$ of diameter $<\gamma$, so that $(\partial \oplus 1_{G'})H$ is geometric over $C^{\varepsilon/2-\gamma} - D^{\varepsilon/2}$.

We now translate this back into geometry. We can stabilize ∂ by adding small cancelling pairs of k, $k+1$ handles. This does not change anything appreciably. We can "deform" a handlebody by adding a parallel copy of one $k+1$ handle to another along an arc between them. By this operation and by changing orientations and choice of lift of handles to the cover $N^\wedge$ we can realize an "elementary matrix" (8.2) on the chain group G_{n+1}. If the algebraic deformation H has diameter δ, then we never alter a handle by anything further away than δ (see Definition 8.2). The diameter of the resulting handledody depends mostly on the arcs along which the addition is performed. Using the local connectedness of X and the $(\delta, 1)$ connectedness of $N \to X$ (hence of $\partial_1 N_h \to X$) we see that if δ is small enough these arcs may be chosen small also. Thus we can change the handlebody to another small one with boundary $(\partial \oplus 1_{G'})H$.

Finally we want to remove excessive geometric intersections. This is usually done with "Whitney trick" which gives an isotopy of $\partial_1 N_k$ constructed by pushing across a 2-disc to remove an algebraically cancelling pair of intersections. The diameter of this isotopy depends on the size of the 2-discs. Using the local 1-connectedness of X, and the $(\delta, 1)$ connectedness

of $N\hat{} \to X$, we see that if δ is small enough these 2-discs can also be chosen small. This argument is where we need $n \geqq 6$. If $n = 5$, then the "stable" Whitney trick of Quinn [3] applies after stabilization by $S^2 \times S^2 \times I$.

Returning to the proof of the h-cobordism theorem, we had found an algebraic δ deformation of $\partial: G_{n+1} \to G_k$ to a homomorphism which over C^ε is induced by a bijection of basis elements of diameter $<\delta$. Supposing that δ is small enough, we can apply the arguments above to get a handlebody of diameter $\gamma < \varepsilon$ such that over C^γ each $k+1$ handle intersects exactly one k handle, in a single disc. The union of such a pair of handles is a disc, intersecting the collar in a disc. Therefore it can be absorbed into the collar. Since these pairs are all disjoint, and of diameter $<2\gamma$, we can absorb them all simultaneously, and obtain a product structure on $(N, \partial_0 N)$ over C, of diameter 3γ.

If $3\gamma < \varepsilon$, this completes the proof of the theorem.

7. Proof of the approximate end theorem

The theorem is that an end with a long enough tameness structure has an approximate completion. This is proved in a sequence of steps which improve the neighborhoods in the structure at the expense of shortening it. There are three stages in the proof. The first involves low dimensional arguments to obtain neighborhoods U_i with $\partial_0 U_i$ $(\delta, 1)$ connected. The second stage uses geometrical connectivity and induction to make $(U_i, \partial_0 U_i)$ relatively $(\delta, n-3)$ connected. The final stage translates relatively $n-3$ connected structures into stable geometric algebra so the theorem of Section 8 can be applied. This translates back into geometric information which is used to complete the proof.

7.1 *The setting.* We have $e: M \to X$, $X \supset C \supset D$, $\varepsilon > 0$ sequences $M \supset U \supset U_1 \supset \cdots \supset U_k \supset W_0 \supset W_1 \supset \cdots \supset W_k$, and a regular π cover $M\hat{} \to M$ satisfying:

1) Each U_i is a manifold neighborhood of the end of e, $U_i \subseteqq U^0_{i-1}$, and the U_i are levels in a collar over D^ε.

2) Over C^ε, $U_j - W_g$ δ deformation retracts in $U_j - W_{j+1}$ into $U_j - U_{j+1}$, rel $\partial_0 U_j$.

3) $U_j\hat{} - W_j\hat{} \to U_{j-1}\hat{} - W_{j+1}\hat{}$ is $(\delta, 1)$ connected over C^ε.

4) $\partial_0 U_j\hat{} \to U_{j-1}\hat{} - U_{j+1}\hat{}$ is $(\delta, 1)$ connected over C^ε.

We are using manifold neighborhoods rather than the open ones of the definition of a tame structure. To get this simply find manifolds between adjacent open neighborhoods and interpolate (see below).

Suppose $U_i \supset V \supset U_{i+1}$. If we delete U_i, U_{i+1}, and W_{i+1} from the tame structure, insert V, and associate it with W_i, the result is a tame structure with length $k-1$. We refer to this process as *interpolation*. Thus if we have a structure, and can use r of the neighborhoods to find a nicer one, we interpolate to get a nicer tame structure with length $1/r$ times the original.

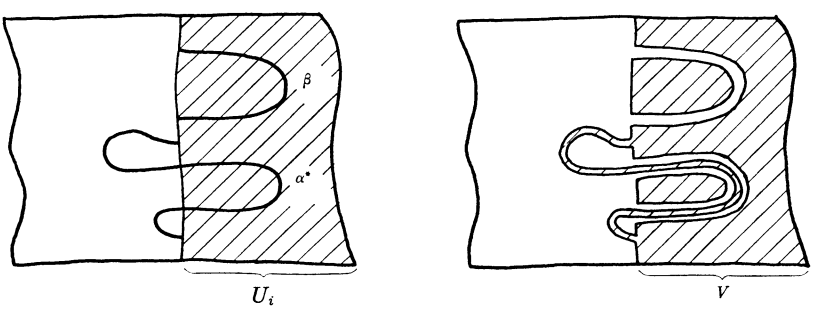
We will not mention the subset D. Our handlebody structures are always collars over D^ε; so as long as the manipulations are of diameter $<\varepsilon$ nothing happens over D.

7.2 $\partial_0 U_i\hat{}(\delta, 1)$ *connected.* First note that $(U_i - W_{i-1}, \partial_0 U_i) \to (U_i - W_i, \partial_0 U_i)$ is relatively $(\delta, 0)$ connected over C^ε (since the deformation (2) for $i-1$ must pass points through $\partial_1 U_i$) and $\partial_0 U_i$ is $(2\delta, -1)$ connected over C^ε (using this and (4)).

We make $\partial_1 U_i$ 0-connected next. Choose $\gamma > 0$ so that points of distance $<\gamma$ in X can be joined by arcs of diameter $<\varepsilon/2$, and cover C^ε by a finite number of open sets $\{S_j\}$ of diameter $<\gamma$. Suppose δ is so small that two points closer than 4δ are both contained in one of these open sets. We can find manifolds $G_j \subset M$ so that $e^{-1}(S_j^{-\delta}) \subset G_j \subset e^{-1}(S_j)$, and are transverse to $\partial_0 U_i$, $\partial_0 U_{i+1}$. For each j choose a set of points $\{\alpha_{j,k}\}$, one in each component of $G_j \cap \partial_0 U_i$, and a finite set of arcs $\{\beta_{j,r}\}$ in $\big((U_i - U_{i+1}^0) \cap G_j, \partial_0 U_i \cap G_j\big)$. The arcs are chosen so that if we kill $\pi_0(\partial_0 U_i \cap G_j)$ by attaching arcs on the points $\{\alpha_{j,k}\}$, then attaching the arcs $\{\beta_{j,r}\}$ will kill the relative π_1 of the result.

Now construct $U_i \supset V' \supset U_{i+1}$ by approximating the arcs $\{\beta_{j,r}\}$ by disjointly embedded 1-handles in $(U_i - U_{i+1}^0, \partial_0 U_{i+1})$, and deleting the interiors from U_i.

Next, since the diameter of $e(\{\alpha_{j,k}\}) < \gamma$, we can join these points by arcs of diameter $<\varepsilon/2$ in X. Since $\partial_1 U_i \to U_{i+1}$ is $(\delta, 1)$ connected (over C^ε) these δ lift to give arcs $\{\alpha_{j,n}^*\}$ in $(U_{i-1} - U_{i+1}) \cap e^{-1}(S_j^\delta)$. They may go back and forth across $\partial_1 U_i$, but we can arrange that they start off going into $U_{i-1} - U_i$. Approximate them by disjointly embedded arcs disjoint from



the 1-handles gotten from the $\{\beta_{j,r}\}$, and transverse to $\partial_1 U_i$. Where an arc goes into U_i, thicken the intersection to a 1-handle and delete it from U_i. This gives V'' disjoint from the arcs. Now thicken the arcs into 1-handles and add to V'' to get V (see the illustration on the previous page).

We claim that $\partial_0 V$ is $(\varepsilon, 0)$ connected over C, and that $(V - W_i, \partial_1 V) \to (V - W_{i-1}, \partial_1 V)$ is relatively $(\varepsilon, 1)$ connected over C.

Suppose we have a relative 1-complex (R, S) and a diagram

$$\begin{array}{ccc} S & \longrightarrow & \partial_1 V \\ \cap & & \downarrow e \\ R & \longrightarrow & C \subseteq X . \end{array}$$

We assume S is a 0-complex. If a point of S falls in one of the new 1-handles of V, it $\varepsilon/2$ deforms off the handle into $\partial_1 V \cap \partial_1 U$. Therefore assume $S \to \partial_1 V \cap \partial_1 U$. Subdivide R into arcs of diameter $<\delta$. We can δ lift the 0 skeleton (by the remark above) to $\partial_1 U_1$ and push into $\partial_1 U \cap \partial_1 V$. By choice of δ the lifts of the endpoints of one of these arcs lie in some $G_j \cap \partial_1 U$. They can therefore be joined in $\partial_1 V \cap e^{-1}(S^\delta)$ using the 1-handles from the arcs $\{\alpha^*_{j,n}\}$. This gives an ε lift of R.

Next suppose $(R, S) \to (V - W_i, \partial_0 V)$ is a relative 1-complex over C^ε. Again push S into $\partial_0 U_i \cap \partial_0 V$. If an arc passes through a 1-handle added to U_i, push that part to the boundary. Then we can add it to S. This gives $(R, S) \to U_i - W_i, \partial_0 U_i \cap \partial_0 V$. Subdivide into arcs of diameter $<\delta$ and deform the 0-skeleton into $\partial_0 U_i \cap \partial_0 V$. Now use (2) to deform the resulting arcs into $U_i - U_{i+1}, \partial_0 U_i - \partial_1 V$. These arcs are of diameter $<3\alpha$, so they lie in some G_j. They therefore ε deform into $\partial_0 V$ because of the 1-handles from arcs $\{\beta_{j,r}\}$. This shows that $(V - W_i, \partial_0 V) \to (V - W_{i+1}, \partial_0 V)$ is relatively $(\varepsilon, 1)$ connected over C.

We are ready to get U_i with $\partial_0 U_i\hat{}\,(\delta, 1)$ connected. We can interpolate neighborhoods from the above argument to get a tame structure with 1/3 the length, but with $\partial_0 U_i$ $(\delta, 0)$ connected and $(U_i - W_i, \partial_0 U_i) \to (U_i - W_{i+1}, \partial_0 U_i)$ relatively $(\delta, 1)$ connected over C^ε. The goal is to use a similar argument adding and deleting 2-handles to construct V.

Cover C^ε by $\{S_j\}$ such that a loop in S_j extends to a disc of diameter $<\varepsilon/2$ and presume δ is so small that a loop of diameter $<4\delta$ lies in one of the S_j. Approximate $e^{-1}(S_j)$ by manifolds G_j, and choose loops $\{\alpha_{j,k}\}$ in $(\partial_0 U_i \cap G_j)\hat{}$ so that they generate the fundamental groups of the components. Span the images $e(\alpha_{j,k})$ by $\varepsilon/2$ discs in X, and use the $(\delta, 1)$ connectedness to lift to discs $(D^2, S^1) \to (U_{i-1}\hat{} - U_{i+1}\hat{}, \partial_0 U_i\hat{})$. We can assume these are disjointly embedded, start out going into $U_{i-1} - U_i$, and intersect $\partial_0 U_i$ transversely.

These discs therefore intersect U_i in surfaces lying in the interior of D^2.

These discs $D^2 \to (U_{i-1} - U_{i+1}, \partial_0 U_i)$ must be modified to intersect U_i in 2-discs before we can construct V. Begin by choosing a system of arcs in each disc $\bigcap U_i$ so that if we cut along these arcs, we get discs. By the previous step $(U_i - W_i, \partial_0 U_i) \to (U_i - W_{i+1}, \partial_0 U_i)$ is relatively $(\delta, 1)$ connected, so we can pull the images of these arcs (which lie in $U_i - U_{i-1}$) into $\partial_0 U_i$ inside $(U_i - W_{i+1})$. Further, by (2) we can pull the homotopy into $U_i - U_{i+2}$. Approximate the homotopy by an embedded 2-disc, and push the arc on the surface along this 2-disc and across $\partial_1 U_i$. This cuts the intersection into 2-discs.

Now proceed as before: delete neighborhoods of the 2-disc intersection with U_i to get V' disjoint from the 2-discs, and then thicken the 2-discs and add to get V. This gives $U_{i-1} \supset V \supset U_{i+2}$. The proof that $\partial_0 V\hat{}$ is $(\varepsilon, 1)$ connected goes as above. We remark only that the reason it was important to delete only 2-handles from U_i is that if S is a 1-complex (coming from (R, S), a relative 2-complex), then we can deform $S \to \partial_0 V$ into $\partial_0 U_i \cap \partial_0 V$.

As usual we change ε, δ, and interpolate to get a tame structure with 1/4 the length and $\partial_1 U_i\hat{}$ $(\delta, 1)$ connected over C^ε.

7.3 $(U_i, \partial_0 U_i)$ *relatively* $n - 3$ *connected.* First note that since the $\partial_0 U_i$ are $(\delta, 1)$ connected over C^ε, $(U_i - U^0_{i+1}, \partial_0 U_i)$ is relatively $(2\delta, 1)$ connected.

Now suppose we have a tameness structure with $(U_i - W_i, \partial_0 U_i) \to (U_i - W_{i+1}, \partial U_i)$ $(\delta, k-1)$ connected over C^ε, $2 \leqq k \leqq n - 3$. Take a handlebody structure on $(U_i - U_{i+1}, \partial_0 U_i)$ of diameter $< \delta$. The $k-1$ skeleton δ deforms into $\partial_0 U_i$ over C^ε, so the k cells give maps $(D^k, S^{k-1}) \to (U_i - W_{i+1}, \partial_0 U_i)$. The standard hypothesis (2) can be used to push these into $(U_i - U_{i+1}, \partial_0 U_i)$, where they have diameter $< 5\delta$.

Our first goal is to deform these discs to embeddings. If $2k < n$ (e.g., if $k = 2$) this is easily done by general position. We have to work for the other cases. Denote $U_i - U^0_{i+1}$ by N.

The connectivity hypotheses imply that $(N, \partial_0 N)$ is relatively $(2\delta, k-2)$ connected over C^ε. Therefore if $\gamma > 0$ and δ is small enough, the geometrical connectivity theorem implies that there is a handlebody structure of diameter $< \gamma$ with no handles of dimension $\leqq k - 2$.

Deform the k-discs into the k skeleton N_k so they are transverse to the k handles, and intersect $N_k - N_{k-1}$ in a collection of discs. Each map $\alpha_i: (D^k, S^{k-1}) \to N_k, \partial_0 N$ may contain several of these intersection discs. Join these by arcs in D^k. Since $k \leqq n - 2$, these arcs deform rel ends into $\partial_1 N_k$, and there we can approximate the maps α_i on neighborhoods of these arcs

by disjoint embeddings. This yields for each i a disc $D_i \subset D^k$ such that $\alpha_i | D_i \to N_k$ int N_{k-1} are disjoint embeddings, and $\alpha_i(D^k - D_i^0) \subset N_{k-1}$.

The map $\alpha_i | (D^k - D_i^0)$ is a homotopy from the embedding $\alpha_i | \partial D_i \subset \partial_1 N_k$, to $\alpha_i | \partial D^k \to \partial_0 N$. We claim that these homotopies can be replaced by disjoint embeddings.

Pass to the cover $N\hat{}$. Since $\partial_0 U_0\hat{}$ is $(\delta, 1)$ connected over C^ε, and $3 \leqq k \leqq n - 3$, $\partial_1 N_{k-1}\hat{}$ is $(3\delta, 1)$ connected over $C^{\varepsilon - 3\delta}$. Consider lifts of the spheres $\alpha_i\hat{}(\partial D_i) \subset \partial_1 N_{k-1}\hat{}$. They are homotopic into $\partial_0 N\hat{}$, so they are zero homologically in $H_{k-1}(N_{k-1}\hat{}, \partial_0 N\hat{})$. But this is the free $\mathbf{Z}\pi$ module generated by the $k - 1$ handles, and the homology class is detected by intersections with the dual cells. Therefore the intersection of each $\alpha_i(\partial D_i)$ with the boundary of the duals of the $k - 1$ handles is algebraically zero.

Therefore if $\rho > 0$ and δ is small enough, there is a ρ isotopy of $\partial_1 N_{k-1}$ which moves these spheres disjoint from these dual handles. They then lie in $\partial_1 N_{-1}$, so the product structure in the collar gives disjoint cylinders descending to $\partial_0 N$. The existence of this isotopy follows from the Whitney trick argument used in the proof of the thin h-cobordism theorem (second paragraph from the end of § 6).

This achieves the first goal, embedding the k-discs. Define V, $U_i \supset V \supset U_{i+1}$, by deleting the interiors of small tubular neighborhoods of these discs. We now show that if δ is small enough, V is (ε, k) connected. Let A denote the union of these tubular neighborhoods, so that $U_i = V \cup A$.

First note that

$$(U_i - W_i, \partial_0 U_i \cup A) \longrightarrow (U_i - W_{i+1}, \partial_0 U_i \cup A)$$

is relatively k connected: a relative k complex deforms into $U_i - U_{i-1}$, then into the k skeleton of $U_i - U_{i-1}$, and thence into A. Now we consider the excision $(U_i - W_i, \partial_0 V) \to (U_i - W_i, \partial_0 U_i - A)$. Composing with the above shows that $(V - W_i, \partial_0 V) \to (V - W_{i+1}, \partial_0 V)$ is homologically relatively homologically k connected. If $W_i = W_{i+1}$ (e.g., if $W = \varnothing$) we could conclude from the eventual Hurewicz theorem that if δ is small enough, $(V - W_i, \partial_0 V)$ is relatively (ε, k) connected over C. In general we apply the Hurewicz theorem to the sequence

$$(V - W_i, \partial_0 V) \longrightarrow (V - W_{i+1}, \partial_0 V) \longrightarrow \cdots \longrightarrow (V - W_{i+j}, \partial_0 V)$$

noting that each of the maps is relatively homologically k trivial.

This step requires 2^{k+2} stages of the tame structure if $W \neq \varnothing$, or 2 if $W = \varnothing$. By induction we therefore obtain a tame structure with neighborhoods

$$(U_i - W_i, \partial_0 U_i) \longrightarrow (U_i - W_{i+1}, \partial_0 U_i)$$

$(\delta, n-3)$ connected, and length reduced by 2^j, $j = n(n-1)/2$ in general, or 2^{n-1} if $W = \varnothing$.

7.4 *Approximate completions.* With the above improvements the setting is complete for the passage to stable algebra.

The manifolds $(U_i - U^0_{i+1}, \partial_0 U_i)$ are relatively $(2\delta, n-4)$ connected over C^ε. Further $(U_i - U_{i+1}, \partial_1 U_i)$ is relatively $(2\delta, 1)$ connected. Therefore if $\gamma > 0$ the geometrical connectivity theorem implies that if δ is small enough there is a handlebody structure on this manifold with diameter $<\gamma$, and handles only in dimensions $n-3$ and $n-2$ over $C^{\varepsilon-\gamma}$.

Let G_i denote the geometric $\mathbf{Z}\pi$ modules corresponding to the i handles, and $\partial: G_{n-2} \to G_{n-3}$ the boundary homomorphism. The first goal is to show we can deform the handlebody structure so that the kernel of ∂ is generated by a set of $n-2$ handles. We begin by showing that $\ker \partial$ is also the kernel of a projection.

As above there is a homotopy

$$H_t: (U_i - U^0_{i+1}, \partial_0 U_i) \longrightarrow (U_i - U_{i+2}, \partial_0 U_i)$$

with $H_0 = I$ and $H_1 = h$, so that h carries the $n-3$ skeleton of $U_i - U^0_{i+1}$ into $\partial_0 U_i$. Let $\partial': G'_{n-2} \to G'_{n-3}$ be the boundary for the handlebody structure on $(U_{i+1} - U^0_{i+2}, \partial_0 U_{i+1})$; then combining the two handlebodies we get a structure on $(U_i - U^0_{i+2}, \partial_0 U_i)$ with boundary

$$\begin{bmatrix} \partial & p \\ 0 & \partial' \end{bmatrix}: G_{n-2} \oplus G'_{n-2} \longrightarrow G_{n-3} \oplus G'_{n-3}\,.$$

The homotopy above gives a chain homotopy

$$\begin{bmatrix} H \\ H' \end{bmatrix}: G_{n-3} \longrightarrow G_{n-2} \oplus G'_{n-2}$$

such that (in particular)

$$\begin{bmatrix} H \\ H' \end{bmatrix} \partial = \begin{bmatrix} I \\ 0 \end{bmatrix} - \begin{bmatrix} h \\ 0 \end{bmatrix}.$$

Therefore $H\partial = I - h$. But $\partial h = 0$, so

$$h - h^2 = (I - h)h = (H\partial)h = H(\partial h) = 0\,.$$

Therefore h is a projection (over C^ε). It also follows that $\ker(I-h) = \ker \partial$, since $\ker(I-h) = \operatorname{im}(h)$.

Therefore $\ker \partial$ is the kernel of the δ projection $(I-h)$ over C^ε. According to 8.4, if $\gamma > 0$ there is a δ so that we can stabilize $(I-h)$ by a geometric projection $q: G' \to G'$ and then find a γ deformation E of $G_{n-2} \oplus G'$ so that the kernel of $(1 - h \oplus q)E$ is a geometric submodule.

We can stabilize h by an identity by adding a cancelling pair of $n-2$, $n-3$ handles with an appropriate homotopy H. We can add a zero map by adding a cancelling pair and then deleting the interior of the $n-3$ handle from U_i. Combining these operations allows realization of the algebraic stabilization.

Next, as in the thin h-cobordism theorem, we can realize the algebraic deformation E by changing lifts and small handle additions. This realizes the first goal, of generating $\ker \partial$ by handles.

The next step is to make the boundary $n-3$ spheres of these handles disjoint from the dual 2 spheres of the $n-3$ handles, in the intermediate level between the handles. The algebraic intersections are zero, but the codimension is too low to apply the Whitney argument directly.

Call the intermediate level W. We have the additional information that $W^{\wedge}$ is obtained from the $(\delta, 1)$ connected manifold $\partial_0 U_{i+1}^{\wedge}$ by adding 2-handles. The connectivity implies these 2 handles are added trivially. Therefore the $n-3$ spheres in W have transverse 2 spheres which intersect in exactly one point. Now we can make the dual 2 spheres disjoint from the $n-3$ spheres by, at each intersection point, forming the connected sum with the transverse 2-sphere. Since the intersections are algebraically zero, the results of this operation are homotopic to the original dual 2-spheres. In $n \geqq 6$ they are therefore isotopic, and if $n=5$ they are after connected sums $W \sharp S^2 \times S^2$ (Quinn [3]). (This is the only use of $n \geqq 6$.)

Now we have $\ker \partial$ generated by $n-2$ handles with boundary on $\partial_0 U_i$. Form V by deleting from U_i the interiors of these $n-2$ handles. It follows from the eventual Hurewicz theorem exactly as in 7.3 that $(V - W_i, \partial_0 V) \to (V - W_{i+j}, \partial_0 V)$ is n connected for large enough j, and therefore an approximate completion. Again this requires n stages in general, but only 2 if $W = \varnothing$.

This completes the proof of the approximate end theorem.

8. The stable algebra of geometric modules

In Sections 6, 7 certain handlebody problems were reduced to "stable algebra" problems on geometric modules. This section provides solutions to these geometric module problems. We will not define K groups of geometric groups, but will prove directly analogues of the vanishing theorem $\mathrm{Wh}(\pi) = 0 = \tilde{K}_0(\pi)$ (for certain π).

The concept of geometric group was introduced by Connell and Hollingsworth in 1966 [1]. Their paper stated some conjectures, and showed that these imply several important topological conjectures. The results of this

section are generalizations of the Connell-Hollingsworth conjectures.

1.8 *Definition.* Suppose X is a metric space, and π is a group.

1) If $\{x_\alpha\}$, $a \in A$, is a collection of points in X, then the *geometric* $\mathbf{Z}\pi$ *module* with basis $\{x_\alpha\}$ is the free module $\mathbf{Z}\pi(\{x_\alpha\})$. The basis set will always be finite unless explicitly said to be infinite.

2) If $h: G_1 \to G_2$ is a homomorphism of geometric modules with bases $\{x_\alpha\}$, $\{y_\beta\}$ respectively, then the *underlying set function* is defined by $\underline{h}(x_\alpha) = \{y_\alpha \mid y_\alpha \text{ has nonzero coefficient in } h(x_\alpha)\}$.

3) The *diameter* of h is $\sup_\alpha$ diameter $(\{x_\alpha\} \cup \underline{h}(x_\alpha))$.

In the applications we have a handlebody mapping to X with the images of handles small. They can reasonably be thought of as being concentrated at points, and these are the bases for the geometric modules. The boundary homomorphism between j and $j-1$ handles is defined by geometric intersection, and so gives a homomorphism of small diameter. The next definition codifies how a geometric module can be "deformed" by deforming the handlebody structure.

8.2 *Definition.* Suppose G is a geometric $\mathbf{Z}\pi$ module on X with basis $\{x_\alpha\}$.

1) A *deformation* of G is a finite composition $H = H_0 H_1 \cdots H_k$ such that each H_i is elementary: $H_i(X_\alpha) = \sum n_{\alpha\beta} x_\beta$ satisfies a) $n_{\alpha\alpha} = \pm g_\alpha$, $g_\alpha \in \pi$ and b) $n_{\alpha\beta} = 0$ for all except possibly one pair (α, β) with $\alpha \neq \beta$.

2) The *diameter* of a deformation is the diameter of the composition of the underlying set functions of the deformation, and its inverse deformation: $\sup_\alpha \operatorname{diam}(\{x_\alpha\} \cup \underline{H}_1 \underline{H}_2 \cdots \underline{H}_k(x_\alpha) \cup \underline{H}_k^{-1} \cdots \underline{H}_1^{-1}(x_\alpha))$.

Finally we define the objects we are interested in deforming. If G is a geometric module on X and $C \subseteq X$, we let $G \mid C$ denote the geometric submodule generated by basis points which are in C. Denote the projection to, and inclusion of, this submodule by p_c and i_c respectively.

8.3 *Definition.* Suppose G_1, G_2 are geometric modules on X, $X \supseteq C$, and $h: G_1 \to G_2$ is a homomorphism.

1) h is a δ *isomorphism over* C if there is a homomorphism g so that both have diameter less than δ, and $p_c g h i_c = p_c h g i_c = I_c$.

2) h is a δ *projection over* C if $G_1 = G_2$, h has diameter less than δ, and $p_c h^2 i_c = p_c h i_c$.

3) An isomorphism or projection will be called *geometric* if (isomorphism case) it is induced by a bijection of basis sets, or (projection case) takes each basis element either to itself or zero.

To be able to deform a δ-isomorphism to the identity is clearly the

analogue of an isomorphism being zero in the Whitehead group. A projection on a free module determines complementary projective summands $\ker h$ and $\operatorname{im} h$. Therefore deforming a δ projection to a geometric projection is analogous to finding a basis for a projective module.

8.4 THEOREM. *Suppose X is a compact metric space, $X \supset C \supset D$ are closed subsets with $D \subseteqq C^0$, and $\varepsilon > 0$. Then there is $\delta > 0$ such that if π is a group with $\operatorname{Wh}(\pi \times \mathbf{Z}^n) = 0$ for all n, the following are true:*

1) *If $h: G_1 \to G_2$ is a δ isomorphism over C, then there is a geometric δ isomorphism g and an ε deformation H such that $H(h \oplus g)$ is a geometric isomorphism over D.*

2) *If $h: G \to G$ is a δ projection over C, then there is a geometric projection g and ε deformations H_1, H_2 such that $H_1(h \oplus g)\ H_2$ is a geometric projection over D.*

We outline the proof briefly. The first step is a simple reduction to the manifold case. Next is an extension lemma on a disc. This is the heart of the proof, and is proved using a torus argument. (See e.g., Lacher [1], § 4.) The next step is to decompose an isomorphism or projection on a disc. This provides the induction step in a handlebody argument which decomposes an isomorphism or projection on a manifold into many small pieces. Finally the hypothesis $\operatorname{Wh}(\pi) = \tilde{K}_0(\mathbf{Z}\pi) = 0$ is used to deform each of these little pieces to the desired form.

Reduction to manifolds. Let I^∞ denote the Hilbert cube with the standard metric, and let $X \subset I^\infty$ be an embedding. We use the induced metric on X.

Suppose $\varepsilon > 0$. Then inside the $\varepsilon/6$ neighborhood of X in I^∞ is a neighborhood of the form $U \times I^{\infty - j}$, where $U \subset I^j$ is a codimension zero submanifold, and the diameter of $I^{\infty - j}$ is $< \varepsilon/6$.

We can pass back and forth from X to U. Since the projection $p: X \to U \times I^{\infty - j} \to U$ decreases distances, a δ homomorphism on G on X yields a δ homomorphism on p_*G on U. Now suppose we have a geometric module G' on U. Since $U \times I^{\infty - j}$ is inside an $\varepsilon/6$ neighborhood we can lift the basis to get $G\hat{}$ on X with $p_*G\hat{}$ within $\varepsilon/6$ of G'. Finally since $I^{\infty - j}$ has diameter less than $\varepsilon/6$, an $\varepsilon/3$ deformation of p_*G is a $2\varepsilon/3$ deformation of G.

We leave it to the reader to see that if ε is small enough, there are submanifolds $U \supset V \supset W$ so that the hypothesis on C projects to the hypothesis on V, and the conclusion of the theorem on W lifts to give the conclusion on D.

The next lemma is the main step in the manifold case. It says roughly

that given a homomorphism on a geometric group on D^n which is a δ automorphism (projection) over $(2/3)D^n$, then the restriction to $(1/2)D^n$ can be extended to an ε automorphism (projection) on all of D^n.

This is not quite true if $u = 1$, since a homomorphism which shifts from one end of D^1 to the other cannot be completed to be an isomorphism. To deal with this we define the "flux" of such a homomorphism: Suppose $h: G \to G$ is a homomorphism of a geometric $\mathbf{Z}\pi$ module over D^1 which is an ε isomorphism over $(2/3)D^1$, for $\varepsilon < 1/6$. Then define $h' = h \otimes_\pi \mathbf{R}$, a homomorphism of a geometric $\mathbf{R}$ module on D^1. h' is still an isomorphism over $(2/3)D^1$, so the kernel splits into pieces $\ker_+ h' \oplus \ker_- h'$, which lie over $(0, 1]$ and $[-1, 0)$ respectively. The image splits similarly so we can define the *flux* of h to be $\dim(\ker_- h') - \dim(\mathrm{im}_- h')$.

This flux problem arises because we are considering automorphisms. It will disappear later when we are able to change the identification of the domain and range.

8.5 LEMMA. *Suppose n is given, π is a group such that $\mathrm{Wh}(\pi \times \mathbf{Z}^i) = 0$ (for all i), and $\varepsilon > 0$. Then there is $\delta > 0$ such that the following are true:*

1) *If G is a geometric $\mathbf{Z}\pi$ module over D^n, $h: G \to G$ a δ automorphism over $(2/3)D^n$ of flux 0 if $n = 1$ then there are modules G', G'' such that $G \oplus G' | (1/2)D^n = G'' | (1/2)D^n$, and an ε automorphism $h: G'' \to G''$ such that on $(1/2)D^n$, $h'' = h \oplus I_{G'}$.*

2) *If G is geometric over D^n, $h: G \to G$ is a δ projection over $(2/3)D^n$, then there are modules G', G'' such that $G \oplus G' | (1/2)D^n = G'' | (1/2)D^n$, an ε projection $h'': G'' \to G''$, and a geometric projection $h': G' \to G'$ such that on $(1/2)D^n$, $h'' = h \oplus h'$.*

Proof. The steps are those of a typical Kirby torus argument: pull up to a punctured torus via an immersion, fill in across the puncture, pass to the universal cover, truncate to get a finite thing, and then project back to the disc.

Think of T^n as $\mathbf{R}^n/2\mathbf{Z}^n$, so $(1/2)D^n \subset T^n$. Let B be a ball disjoint from $(1/2)D^n$, and let $j: T^n - \mathrm{int}(1/2)B \to (2/3)D^n$ be an immersion which is the identity on $(1/2)D^n$.

If G is a geometric module on D^n, with basis $\{x_\alpha\}$, define j^*G to be the geometric module with basis $j^{-1}(\{x_\alpha\})$. Suppose $\gamma > 0$, and is smaller than the minimum distance between distinct point inverses in $T^n - (1/2)B$. Then there is $\delta > 0$ such that if $y \in T^n - (1/2 - \gamma)B$ and $x \in D^n$ with $d(j(y), x) < \delta$, then there is a unique $x' \in T^n - (1/2)B$ such that $d(y, x') < \gamma$, and $j(x') = x$.

A homomorphism of diameter $<\delta$ on G on D'' lifts to a homomorphism of diameter $<\gamma$ on j^*G, in a well defined way except perhaps in $(1/2 + \gamma)B$. In particular if h is a δ automorphism (projection) over $(2/2)D^n$, then j^*h is a γ automorphism (respectively projection) over $T^n - (1/2 + \gamma)B$.

Next we claim that there is a geometric module G' on $(1/2)B$, and a homomorphism $h': G \oplus G' \to G \oplus G'$ which is an automorphism (or a projection) so that h' restricted to $T^n - (2/3)B^n$ is equal to h, and $h'(G \oplus G' \mid (2/3)B^n) \subset G \oplus G' \mid B^n$.

We pause for some notation. Let

$$G_0 = G\Big|\Big(\frac{1}{2} + \delta\Big)B - \frac{1}{2}B\,, \qquad G_1 = G\Big|\Big(\frac{1}{2} + 3\delta\Big)B - \Big(\frac{1}{2} + \delta B\Big),$$

and

$$G_2 = G \mid T^n - \Big(\frac{1}{2} + 3\delta\Big)B\,.$$

i_r, p_r denote inclusion, resp. projection to G_r, and G_{12} means $G_1 \oplus G_2$, etc. The automorphism and projection cases are separated now.

Automorphisms. We are given a γ homomorphism $g: G \to G$ such that $p_{12}ghi_{12} = I_{G_{12}}$. It follows that $hi_{12}p_{12}g$ is the projection of G onto the (free) direct summand $(\operatorname{im} hi_{12}) \subset G$. Denote the complementary projection $1 - hi_{12}p_{12}g$ by q. Then

$$\begin{bmatrix} hi_{12} & q \\ 0 & p_{12}q \end{bmatrix}: G_{12} \oplus G \longrightarrow G \oplus G_{12}$$

is an isomorphism.

If we rewrite the first module as $G_{12} \oplus G = G_{12} \oplus (G_0 \oplus G_{12}) = G \oplus G_{12}$, then in the claim, $G' = G_{12}$, and h' is this homomorphism. The only part of the claim which needs to be checked is that the image of $G_0 \oplus G'$ lies inside $G_{01} \oplus G'$. $G_0 \oplus G'$ is the G in the domain of the construction above, and the image of this in the G in the range is $\operatorname{im} q$ = image $(1 - hi_{12}p_{12}q)$. This will be in G_{01} if p_2 applied to it is zero. To see this, first observe that $p_2 = p_2p_1 = p_2hgi_{12}p_{12} = p_2hg$. The last step holds because $\operatorname{im}(1 - i_{12}p_{12}) \subset G_0$, hg is a 2γ homomorphism, so $\operatorname{im}\big(hg(1 - i_{12}p_{12})\big) \subseteq G_0 \oplus G_1$, and the projection p_2 takes this to zero. For a similar reason, $p_2hg = p_2hi_{12}p_{12}g$. Therefore $p_2(1 - hi_{12}p_{12}g) = 0$.

Projections. This case is a little less tidy. Essentially what we do is to reduce it to the isomorphism case over $\pi \times \mathbf{Z}$, analogous to the standard inclusion $\tilde{K}_0(\mathbf{Z}\pi) \subset \mathrm{Wh}(\pi \times \mathbf{Z})$.

Additional notation follows: T is an infinite cyclic group with generator t. If G is a $\mathbf{Z}\pi$ module, $G[T]$ is the $\mathbf{Z}[\pi \times T]$ module $G \otimes \mathbf{Z}[T]$. There is a natural inclusion $G \subseteq G[T]$, and we define $G[t^k, t^{k+r}] = \sum_{i=0}^{r} t^{k+i}G$.

Now let $h: G \to G$ be the γ projection above. Define $H = h[T] + t(1-h)[T]: G[T] \to G[T]$; then this is a γ isomorphism over $T^n - (1/2 + \gamma)B$. (The inverse is $J = h[T] + t^{-1}(1-h)[T]$.) Denote the projection $p_{12}[T]$ as P_{12}, $i_{12}[T] = I_{12}$, etc. Then as above there is an isomorphism

$$\begin{bmatrix} HI_{12} & Q \\ 0 & P_{12}J \end{bmatrix}: G_{12}[T] \oplus G[T] \longrightarrow G[T] \oplus G_{12}[T] ,$$

so that the image of Q lies in $G_{01}[T]$. This implies that there is an s such that

$$(1) \qquad \begin{bmatrix} HI_{12} & Q \\ 0 & P_{12}J \end{bmatrix}\big((G_{12} \oplus G)[t^0, t^\infty)\big) \supseteq G_2[t, t^\infty) \oplus (G_{01} \oplus G_{12})[t^s, t^\infty) ,$$

and

$$(2) \qquad \begin{bmatrix} HI_{12} & Q \\ 0 & P_{12}J \end{bmatrix}\big((G_{12} \oplus G)(t^{-\infty}, t^{-1}]\big) \supseteq G_2(t^{-\infty}, t^{-1}] \oplus (G_{01} \oplus G_{12})(t^{-\infty}, t^{-s}] .$$

Since $(G_{12} \oplus G)[t^0, t^\infty)$ and $(G_{12} \oplus G)(t^{-\infty}, t^{-1}]$ are complementary summands, the inclusions imply that the projections of the images (1) and (2) in $G_2 \oplus (G_{01} + G_{12})[t^{-s}, t^s]$ are complementary summands also. The projection of $G_2 \oplus (G_{01} \oplus G_{12})[t^{-s}, t^s]$ onto the image (1) is the desired projection which extends h on G_2.

To obtain a projection or automorphism on T^n simply consider the free module which is added to G as a geometric module based at the center of B.

We improve this by composing with a map $r: T^n \to T^n$ which is the identity on $(1/2)D^n$, and takes B to its center $0B$. If B is chosen with any care at all, we can find an r which increases distances by less than a factor of 2. The result of the composition is therefore a 2γ automorphism (projection) on T^n which is h on $(1/2)D^n$. Denote this by $h\hat{}: G\hat{} \to G\hat{}$.

Let $p: \mathbf{R}^n \to T^n$ be the universal cover. As above if 2γ is small enough, we can pull $h\hat{}$ back uniquely to a 2γ homomorphism $p^*h\hat{}: p^*G\hat{} \to p^*G\hat{}$. $p^*G\hat{}$ is an infinite geometric $\mathbf{Z}\pi$ module on $\mathbf{R}^n$, but the covering translations give it the structure of a finitely generated free $\mathbf{Z}[\pi \times \mathbf{Z}^n]$ module, and $p^*h\hat{}$ is a $\mathbf{Z}[\pi \times \mathbf{Z}^n]$ homomorphism.

We have finally come to the point where the Whitehead group hypothesis is used. In the isomorphism case $p^*h\hat{}$ is an automorphism, of a finitely generated free based $Z[\pi \times \mathbf{Z}^n]$ module. Since $\mathrm{Wh}(\pi \times \mathbf{Z}^n) = 0$, $p^*h\hat{}$ can be stabilized by direct sum with an identity matrix, and then be

expressed as a product of elementary matrices. In the projection case there are products of elementary matrices H_0, H_1 such that (stably again) $H_0 p^* h \hat{} H_1$ is of the form

$$\begin{bmatrix} I & 0 \\ 0 & 0 \end{bmatrix}.$$

We can stabilize by adding to the geometric module $G\hat{}$ at the point $0B$.

There is no δ control on these elementary matrices (=deformations), but they are bounded when considered as homomorphisms on (infinite) $\mathbf{Z}\pi$ geometric modules on $\mathbf{R}^n$. To get back to $(2/3)D^n$ we will compose with a radial retraction $q\colon \mathbf{R}^n \to (2/3)D^n$ which is the identity on $(1/2)D^n$. We regain control by the shrinkage involved in q, and regain finiteness by "truncation".

To describe the truncation procedure, let $E_1 E_2 \cdots E_m$ be the product of elementary $\mathbf{Z}[\pi \times \mathbf{Z}^k]$ matrices. Our definition of "elementary" allows elements of $\pm\pi \times \mathbf{Z}^k$ on the diagonal. We can rewrite the product so that E_1 is diagonal with entries from $\mathbf{Z}^k$, and E_j is $\mathbf{Z}[\pi]$ elementary for $j > 1$. The $\mathbf{Z}[\pi]$ elementary factors are upper or lower triangular as homomorphisms of infinite $\mathbf{Z}[\pi]$ modules over $\mathbf{R}^n$, so they remain isomorphisms if we simply define E_j' to be the identity over the disc sD^n, and E_j outside. The radius s is chosen large enough so that $q_* E_j'$ has diameter $< \gamma/m$.

Now consider the E_1 term. (I am indebted to F.T. Farrell for pointing out that this must be done separately.) E_1 acts by translating basis elements (by the standard action of $\mathbf{Z}^n$ on $\mathbf{R}^n$). Therefore the module decomposes into "orbits", infinite geometric $\mathbf{Z}\pi$ summands whose basis elements lie along a line in $\mathbf{R}^n$ and on which E_1 acts by shifting each basis element to the next in one direction along the line. In $n > 1$ we can alter one of these to be the identity on a disc as follows: choose a large disc sD^n, and a path on the boundary between the two points of intersection with the line. Delete basis elements inside the disc, and add new ones spaced along the path on the boundary. The homomorphism is extended to the new generators by just shifting in the appropriate direction. As above, the diameter of the disc and the spacing of the new generators is chosen so that the result is small after applying q_*.

In the case $n = 1$, E_1 shifts some orbits toward $+\infty$, some toward $-\infty$. If we have one of each we can splice them together to obtain the identity on a disc sD^1; redefine the image of the last generator shifting down from $+\infty$ to be the first generator in the other orbit going out to $+\infty$. Do the same for $-\infty$, and replace by the identity over sD^1. It is easily seen that the flux of the homomorphism is the difference between the number of

orbits shifted in the two directions. Thus if the flux is zero we can splice in pairs and modify all of E_1 to be the identity on a disc.

In the isomorphism case, suppose $E_1 \cdots E_m p^* h^\wedge = I$. The modified isomorphism E_i' agrees with E_i off a huge disc, so there $E_1' \cdots E_m' p^* h^\wedge$ is still the identity. (Note also that the flux of E_1 is minus that of h in case $n = 1$, so under the hypotheses on h, E_1' is defined.) Delete all generators outside that disc to obtain an isomorphism on a finitely generated geometric module. When we apply q_* to this, $q_*(E_1' \cdots E_m')$ becomes a $2m(\gamma/m) = 2\gamma$ homomorphism, which is the identity on $(1/2)D^n$. Therefore q_* of the truncated $E_1' \cdots E_m' p^* h^\wedge$ is a 6γ isomorphism on $(2/3)D^n$ which agrees with h on $(1/2)D^n$. If we suppose $6\gamma < \varepsilon$, this satisfies the conditions of the lemma.

The projection case is finished in the same way, except that there is a product of elementary matrices on each side of $p^* h^\wedge$. In the $n = 1$ case, the flux problem is avoided by multiplying by a homomorphism of arbitrary flux on a part of the module which is killed (or missed) by the projection.

This completes the proof of 8.5. The next step is to use this to split off a piece of an automorphism or projection. If $X \supset Y$, "stabilization on Y" means direct sum with the identity or a geometric projection on a geometric module on Y.

8.6 LEMMA. *Given n, $\varepsilon > 0$, and π as above, then there is a $\delta > 0$ with the following property: Suppose G is a geometric $\mathbf{Z}\pi$ module on D^n and $h: G \to G$ is a δ automorphism (projection) over $(3/4)D^n$, which has flux 0 if $n = 1$. Then after stabilization on $(2/3)D^n$ there is an ε deformation of h to a sum $h_1 \oplus h_2$ on a module $G_1 \oplus G_2$ with $G_1 | (1/3)D^n = 0$, $G_2 | (D^n - (2/3)D^n) = 0$.*

The point is that the h_2 piece is globally an automorphism (projection) over D^n and so can be considered separately, while h_1 agrees with the original near ∂D^n but has strictly smaller support. This will be used essentially inductively, at each step reducing the support of h_1 by splitting off an h_2 piece.

Proof. First notice that if $A: G \to G$ is a δ homomorphism, then $\begin{bmatrix} 1 & A \\ 0 & 1 \end{bmatrix}$ is a δ deformation of $G \oplus G$. Also note that the composition of a δ_1 and a δ_2 deformation is a $2(\delta_1 + \delta_2)$ deformation.

We begin with projections for a change. If $A: G \to G$ is a projection (i.e., $A^2 = A$), then there is a matrix identity on $G \oplus G$,

$$\begin{bmatrix} I & 0 \\ -A & I \end{bmatrix} \begin{bmatrix} I & I \\ 0 & I \end{bmatrix} \begin{bmatrix} I - A & 0 \\ 0 & A \end{bmatrix} \begin{bmatrix} I & 0 \\ I & I \end{bmatrix} \begin{bmatrix} I & -A \\ 0 & I \end{bmatrix} = \begin{bmatrix} I & 0 \\ 0 & 0 \end{bmatrix}.$$

If A is a δ projection, this defines δ deformations H_1, H_2 so that

$$H_1\begin{bmatrix} I-A & 0 \\ 0 & A \end{bmatrix}H_2$$

is a geometric projection.

Now suppose G is a geometric module on D^n, and $h: G \to G$ is a δ projection over $(3/4)D^n$. If δ is small enough we can apply 8.5 to obtain $h': G' \to G'$, $h'': G'' \to G''$ such that h'' is a γ projection, h' is geometric, and $h'' = h \oplus h'$ over $(1/2)D^n$. Further, both G', G'' are supported on $(2/3)D^n$.

We begin verification of the statement: stabilize h by adding the projection $\begin{bmatrix} I & 0 \\ 0 & 0 \end{bmatrix}$ on $G'' \oplus G''$. As above, this γ deforms to $\begin{bmatrix} h'' & 0 \\ 0 & I-h'' \end{bmatrix} \oplus h$. Over $(1/2)D^n$ this is $h'' \oplus (I-h') \oplus (I-h) \oplus h$. Apply the γ deformation above to the last two factors over $(1/2)D^n$ (extend the deformations by the identity outside $(1/2)D^n$); then on $(1/3)D^n$ this becomes $h'' \oplus (I-h') + \begin{bmatrix} I & 0 \\ 0 & 0 \end{bmatrix}$, the last defined on $G \oplus G \,|\, (1/3)D^n$. These last two pieces are geometric projections, so are direct sums of their restrictions to complementary geometric submodules.

Therefore the lemma is satisfied if $4\gamma < \varepsilon$, and in $G'' \oplus G'' \oplus G$ we set $G_1 = 0 \oplus (G'' \oplus G \,|\, D^n - (1/3)D^n)$, and $G_2 = G'' \oplus (G'' \oplus G \,|\, (1/3)D^n)$.

The proof in the isomorphism case follows the same outline, using the identity (for invertible A):

$$\begin{bmatrix} I & -A \\ 0 & I \end{bmatrix}\begin{bmatrix} I & 0 \\ A^{-1} & I \end{bmatrix}\begin{bmatrix} I & -A \\ 0 & I \end{bmatrix}\begin{bmatrix} I & I \\ 0 & I \end{bmatrix}\begin{bmatrix} I & 0 \\ -I & I \end{bmatrix}\begin{bmatrix} I & I \\ 0 & I \end{bmatrix} = \begin{bmatrix} A & 0 \\ 0 & A^{-1} \end{bmatrix}.$$

This is interpreted as an 8δ deformation from $\begin{bmatrix} I & 0 \\ 0 & I \end{bmatrix}$ to $\begin{bmatrix} A & 0 \\ 0 & A^{-1} \end{bmatrix}$. As above we use this to introduce $\begin{bmatrix} A & 0 \\ 0 & A^{-1} \end{bmatrix}$, and then use A^{-1} to cancel the given isomorphism on $(1/3)D^n$.

This completes the proof of Lemma 8.6, so we are ready for the proof of the main theorem (in the manifold case). Since the lemma applies to automorphisms, the first step is to identify the domain and range of an isomorphism.

8.7 PROPOSITION. *Suppose G_1, G_2 are geometric $\mathbf{Z}\pi$ modules on a connected metric space X. If they have the same rank, and if $\delta > 0$, then there is a geometric module G_3 and a geometric δ isomorphism $G_1 \oplus G_3 \simeq G_2 \oplus G_3$.*

Proof. It is sufficient to prove this in case both ranks are 1, so $G_1 = \mathbf{Z}\pi(x)$, $G_2 = \mathbf{Z}\pi(y)$. Since X is connected there is a sequence $x_0, \cdots, x_n$ such that $d(x_i, x_{i+1}) < \delta$, $x_0 = x$, $x_n = y$. Let $G_3 = \mathbf{Z}\pi(x_1, \cdots x_{n-1})$. Then $G_1 \oplus G_3 =$

$\mathbf{Z}\pi(x_0, \cdots, x_{n-1})$, $G_2 \oplus G_3 = \mathbf{Z}\pi(x_1, \cdots, x_n)$, and the function which shifts the index up by 1 induces the desired isomorphism.

In the main theorem if $h: G_1 \to G_2$ is globally an isomorphism (e.g., $X = C$) the ranks are the same and the lemma applies. If $X \neq C$ then we can add some generators on one side or the other side in $X - C$ without disturbing anything, and make the ranks equal.

Now suppose that $X \supset C \supset D$ are compact n-manifolds, D in the interior of C, and $\varepsilon > 0$. Reduce ε if necessary, so that the ε neighborhood of D is in C. To a first approximation the proof proceeds as follows: Choose a handlebody decomposition for $(D, D \cap \partial X)$ with handles and collar of diameter less that $\varepsilon/4$. Let X_j be the union of the collar on $D \cap \partial X$, union the handles of index $\leqq j$. Parameterize the j handles by $\left(D^j \times (1/3)D^{n-j}, S^{j-1} \times (1/3)D^{n-j}\right) \xrightarrow{\subset} (X_j, X_{j-1})$, and extend these to disjoint embeddings $(D^j \times D^{n-j}, S^{j-1} \times D^{n-j}) \xrightarrow{\subset} (X, X_{j-1})$ of diameter less than $\varepsilon/2$.

Now suppose G is a geometric group on X, and $h: G \to G$ is a δ automorphism (projection) over C. We can assume that $G \mid X_{-1} = 0$ (X_{-1} is the collar on $D \cap \partial X$) by choosing the collar thin and pushing G out by a preliminary deformation. Suppose in fact that $G \mid X_{j-1} = 0$ for some j. If δ is small enough, the restriction to a j handle $h \mid D^j \times D^{n-j}$ is a δ automorphism (projection) over $D^j \times (3/4)D^{n-j}$. Project onto the D^{n-j} factor, and presume that δ is small enough to apply Lemma 8.6. This gives, after stabilizing and deforming, a decomposition $h_1 \oplus h_2$ where $h_1 = h$ on $D^{n-j} - (2/3)D^{n-j}$ and is zero on $(1/3)D^{n-j}$, and h_2 is an automorphism (projection).

Pull back to $D^j \times D^{n-j}$. h_1 patches together with h on the complement of the j handles to give an automorphism (projection) on a module which is zero on X_j. (This is a weak point however, since we have lost δ control in the D^{n-j} coordinate.) The h_2 pieces give an automorphism (projection) on a module on each handle.

Continuing inductively, when j reaches n we find that (after stabilizations and small deformations) h has been split into a piece which is zero on D, and small pieces lying over the handles. Since $\mathrm{Wh}(\pi) = 0$ $(\tilde{K}_0(\mathbf{Z}\pi) = 0)$, each of the small pieces deforms to the identity (a geometric projection). Since the handles are small this is also a small deformation. This provides the deformation required for the theorem.

The problem was that in applying 8.6 to the image on D^{n-j} and pulling back to $D^j \times D^{n-j}$ we lost δ control in the D^j coordinate. Therefore we may not be able to apply 8.6 to the $j + 1$ handles. We also neglected the flux problem on the $n - 1$ handles. We circumvent the first difficulty by

subdividing carefully.

Begin with a handle decomposition of diameter $<\varepsilon/2$ as above. If h_n is 0 on X_{n-1}, then it is already a sum of pieces supported on the n handles. For the next step down choose δ_{n-1} so that if h_{n-1} is a δ_{n-1} automorphism (projection) then the lemma applies to h_{n-1} restricted to the $n-1$ handles $D^{n-1} \times D^1$, and projected to D^1.

The third step is typical of the rest. For each $n-2$ handle $D^{n-2} \times D^2$ subdivide D^{n-2} into handles σ_i so that each projection $\sigma_i \times D^2 \to D^2$ changes distances by less than $(1/2)\delta_{n-1}$. Choose δ_{n-2} so small that if h_{n-2} were a δ_{n-2} automorphism (projection) which was 0 on the new X_{n-2}, the lemma would apply to the restriction of h_{n-2} to each $n-2$ handle projected to D^2, to deform it into pieces by a $(1/2)\delta_{n-2}$ deformation. It follows that the h_{n-1} we get by projecting, deforming, and pulling back is a δ_{n-1} automorphism (projection), and the previous step will apply.

Continue in this way, alternately subdividing the j skeleton into pieces thinner than $(1/2)\delta_{j+1}$, and choosing δ_j so small that the deformation on the j handles yields a δ_{j+1} isomorphism (projection) on the $j+1$ handles. This produces a handlebody decomposition, and $\delta = \delta_0$ so that the previous argument will work.

This completes the proof in the projection case, and in the isomorphism case up to the $n-1$ handles. There, a flux problem arises.

Suppose then that the process is complete over $U \subset X$, which consists of the $n-2$ handles and $n-1$ handles with zero flux. Consider an $n-1$ handle not in U. For the first case suppose there is a circle in $X - U$ which intersects the handle in exactly one point. Then there are small geometric isomorphisms with arbitrary flux through the handle obtained by closely spacing generators along the circle and shifting one way or the other. Cancel the flux by stabilizing by one of these.

In the second case suppose there is a path (in $X - U$ with ends in $X - D$) which intersects the handle in a single point. Again we can cancel the flux by adding a shift along the path. This will fail to be an isomorphism at the endpoints, but this is acceptable since they lie in $X - D$.

Finally if neither of these holds, then the handle separates $X - U$ and the component on one side is disjoint from $X - D$. The homomorphism h is therefore an isomorphism over this component, and the flux is automatically zero. So in any case we can arrange the flux to be zero, complete the process on the handle, and add it to U.

This completes the proof of Theorem 8.4.

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